PET ADOPTION & CARE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the requirements for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "PET ADOPTION & CARE MANAGEMENT SYSTEM" is the bonafide work of NAKSHATH N(241501126), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

The work reported in this project was carried out by the candidate and the application described in this document was designed, configured and tested by her in Zoho Creator. The report has been prepared in accordance with the prescribed format of the Department of Artificial Intelligence and Machine Learning, Rajalakshmi Engineering College.

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DECLARATION

I, NAKSHATH N (241501126), hereby declare that the project report titled “Pet Adoption & Care Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the work carried out by me. The application was designed, configured and tested by me on the Zoho Creator platform using forms, lookup fields, reports, Kanban views, pages, Deluge scripts, scheduled tasks and blueprint workflows.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Zoho Creator documentation and the assistance used during development are acknowledged in this report. The implementation details, test observations and limitations presented in the following chapters correspond to the application state described in Chapter 5.

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ACKNOWLEDGEMENT

I express my sincere gratitude to the management of Rajalakshmi Engineering College for providing the infrastructure and learning environment required to complete this project. I thank the Principal of Rajalakshmi Engineering College for encouraging students to develop practical applications and to apply the concepts learned in the curriculum.

I extend my sincere thanks to Dr. M. Ayyadurai, Head of the Department of Artificial Intelligence and Machine Learning, for the constant support, guidance and encouragement provided throughout the course. I am deeply grateful to C.Arulmozhi, Assistant Professor, Department of Artificial Intelligence and Machine Learning, for the valuable instruction, timely feedback and technical suggestions that shaped the Pet Adoption & Care Management System.

I acknowledge Zoho Corporation for providing the Zoho Creator platform and its documentation, which served as the primary technical reference while configuring forms, lookup fields, reports, blueprints and Deluge scripts. I also thank my classmates who reviewed the application during demonstration practice and offered useful observations on the adoption and care screens.

Finally, I thank my family and friends for their patience and support during the preparation of this report.

NAKSHATH N

ABSTRACT

The Pet Adoption & Care Management System is a Zoho Creator application designed to record pets available for adoption, adopter details, adoption applications, adoption outcomes and continuing care activities in a single low-code environment. The application contains six core forms: Adopter Registration, Pet Registration, Adoption Application, Adoption Record, Pet Health and Care Records, and FollowUp and Care Schedule. Lookup fields connect the forms so that an application refers to an existing adopter and an existing pet rather than repeating their details, and an adoption record refers to the application that produced it.

The system uses list reports and Kanban views to present the stored records, a dashboard page to summarise adoption fees, care costs and average pet weight, and Deluge form workflows to validate the adopter verification status and the pet availability status before an application is saved. Two blueprints model the adoption lifecycle and the post-adoption care lifecycle, and a scheduled task is configured to identify care records whose next due date falls within seven days. During testing, the validation workflows blocked unverified adopters and unavailable pets, the confirmation email was received at the recipient address, and the saved records appeared in the corresponding reports and dashboard counts.

The report documents the configured components separately from the behaviour that still requires verification with separate role-based accounts, and it records the defects found during configuration and the corrections applied to the scripts and the schedule.

Keywords: Low-code development, Zoho Creator, pet adoption management, animal care records, lookup fields, Deluge scripting, workflow automation, blueprint, role-based access.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Pet Adoption & Care Management System
Application Domain	Animal welfare – pet adoption and post-adoption care
Student Name & Register No.	NAKSHATH N (241501126)
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Platform	Zoho Creator (low-code application platform)
Application Name	Pet Adoption Care System
Published / Live URL	Development application used for testing; publication to be confirmed before demonstration
Number of Forms / Subforms	6 forms / no subform used
Number of Reports	Twelve reports: six all-record lists and six Kanban-style status views
Number of Pages / Dashboards	One dashboard page: Pet Care Dashboard
Number of Workflows	4 form workflows: vaccination status prompt, adopter verification check, pet availability check and adoption confirmation email
Scheduled Tasks	1 scheduled task: vaccination due-date reminder (seven-day window)
Blueprints	2 blueprints: Adoption Workflow and Post-Adoption Care Workflow
Deluge Functions	check_adopter_verification; check_pet_availability
Integrations	Zoho Creator email notification action; no external REST API integration
Roles & Profiles Defined	Administrator / Adoption Coordinator and Care Staff
Environments Used	Development environment in Zoho Creator
Date of Deployment	October 2026

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LIST OF ABBREVIATIONS

AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
ER	Entity–Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho AI assistant available in the Creator builder

CHAPTER 1

INTRODUCTION

1.1 Background

Animal shelters, rescue groups and campus animal-welfare clubs handle a steady flow of pets that need to be listed for adoption, evaluated by prospective adopters and supported after they leave the shelter. The records involved are not complicated in themselves – a pet profile, an adopter profile, an application, a decision, a fee and a care history – but they are closely related, and the relationships are what make the process hard to run from paper registers or from a set of independent spreadsheets.

In a typical shelter week a coordinator has to answer questions such as: which pets are currently available, has this adopter been verified, which applications are still waiting for a decision, was the adoption fee paid, and which vaccination is due next week. When each of these questions is answered from a different file, the answers are frequently out of date, and the same pet may be promised to two adopters because the first application was never closed.

The Pet Adoption & Care Management System was developed on Zoho Creator to replace that scattered arrangement with one structured application. The system records pet and adopter details, adoption applications, adoption outcomes, health and care events and follow-up schedules as related records, presents them through list reports, Kanban views and a dashboard, and uses Deluge workflows to stop an application from being submitted for an unverified adopter or an unavailable pet.

Zoho Creator was selected because it offers form design, lookup relationships, report and Kanban configuration, page building, role and permission settings, blueprints and a scripting language in one hosted environment. This allowed the effort to be concentrated on the data model and the validation logic rather than on server setup, which is appropriate for an individual course project.

1.2 Problem Statement

Shelter and club adoption work currently lacks a single, structured process for maintaining pet availability, adopter verification, adoption applications, adoption outcomes and continuing care activities. Because these records are kept in separate

lists, the coordinator cannot easily answer basic questions: is this pet still available, is this adopter verified, what decision was recorded on this application, was the fee received, and which care activity is due next. The specific difficulties observed are listed below.

- Pet details, adopter details and the adoption outcome are maintained in different files, so the same pet or adopter appears with inconsistent names, breed descriptions and contact numbers.
- Availability has to be confirmed by asking the shelter staff before every application is reviewed, because the pet record does not carry a reliable status.
- Adopter verification is remembered rather than recorded, so an application from an unverified or previously rejected adopter is accepted and reviewed again.
- Fee, payment status and the signed agreement are tracked outside the application record, which makes reconciliation at the end of the month a manual exercise.
- Vaccination and follow-up dates sit in a diary or a separate calendar, and a due date is missed whenever the diary is not consulted.
- A consolidated view of how many pets are available, how many applications are pending and how much fee has been collected has to be prepared by hand for every review meeting.

1.3 Objectives

- To design structured Zoho Creator forms for adopter registration, pet registration, adoption applications, adoption records, pet health and care records, and follow-up care schedules.
- To relate the six forms through lookup fields so that an application, an adoption record and a care activity refer to existing pet and adopter records instead of duplicating their details.
- To validate the adopter verification status and the pet availability status with Deluge form workflows before an adoption application is saved.
- To record the adoption outcome together with the adoption fee, the payment status, the agreement and the confirmation state.

- To maintain a pet health and care history that carries the vaccination name, the care status and the next due date, and to track follow-up activities with an assigned person, a priority and a completion state.
- To provide list reports, Kanban status views and a dashboard page that present the same records to the coordinator.
- To define the adoption lifecycle and the post-adoption care lifecycle as blueprints and to configure an email notification for the adoption decision.
- To test the core validations and the access configuration, and to document the observed results, the limitations and the corrections that remain.

1.4 Scope of the Project

The scope of the project covers the six core forms with their field definitions and lookup relationships, the twelve reports made up of six all-record lists and six Kanban views, the Pet Care Dashboard page, the four form workflows with their supporting Deluge code, the vaccination due-date reminder schedule and the two blueprints. The system is intended for the adoption coordinator of a shelter or an animal-welfare club and for the care staff who record post-adoption activities.

The following items are outside the verified scope of this version: an external REST API integration, a separately developed mobile application, online payment gateway integration, digital agreement signing, biometric or photo-based pet identification, and predictive analytics on adoption likelihood. Record-level isolation between the coordinator and the care staff was configured in principle but has to be confirmed with separate test logins, and the application has not been subjected to a formal load test or to a measured production deployment.

1.5 Justification for a Low-Code Approach

A low-code platform supplies the data store, the form renderer, the report engine and authentication as platform services, and the developer describes the application through visual models and configuration. For a shelter adoption workflow this is a good fit: the number of entities is small, the rules are well understood, and the requirement changes as new species, breeds and care protocols are added. Rebuilding the same functionality with a conventional web framework would require a database

schema, server-side validation, session handling, hosting and deployment work, none of which is the learning objective of the course.

Zoho Creator additionally allows the configured behaviour to be extended with Deluge where visual configuration is not sufficient, which is used in this project to read the adopter and pet records and to cancel an invalid submission. The platform documentation describes forms as configurable data-capture components [1], lookup fields as the mechanism for relating records [2], workflows as event-driven automation [3] and list reports as the presentation of stored records [4]. Blueprints, which model permitted stage transitions, are documented separately [5]. These five features are the basis of the application.

1.6 Organisation of the Report

Chapter 1 introduces the problem and the objectives. Chapter 2 studies the existing manual process, compares low-code and traditional development and reviews the platform documentation. Chapter 3 records the stakeholders, the functional and non-functional requirements, the hardware and software requirements and the use case model. Chapter 4 presents the system design, including the architecture, the data model, the form specifications, the workflow design, the navigation design and the access-control design. Chapter 5 describes the implementation on Zoho Creator. Chapter 6 presents the testing strategy, the test cases, the debugging observations and the user acceptance feedback. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 evaluates the results against the objectives and states the limitations. Chapter 9 concludes the report and lists the future enhancements. The appendices record the feature coverage checklist, the course outcome mapping, the Deluge source code, the user manual, the evaluation rubric and the viva-voce record sheet.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing process was studied by following how a small shelter and a campus animal-welfare club handle an adoption from intake to follow-up. The process begins when a pet is taken in, continues with enquiries and applications, and ends with the adopted pet being monitored through vaccination and care reminders. The steps are shown in Figure 2.1.

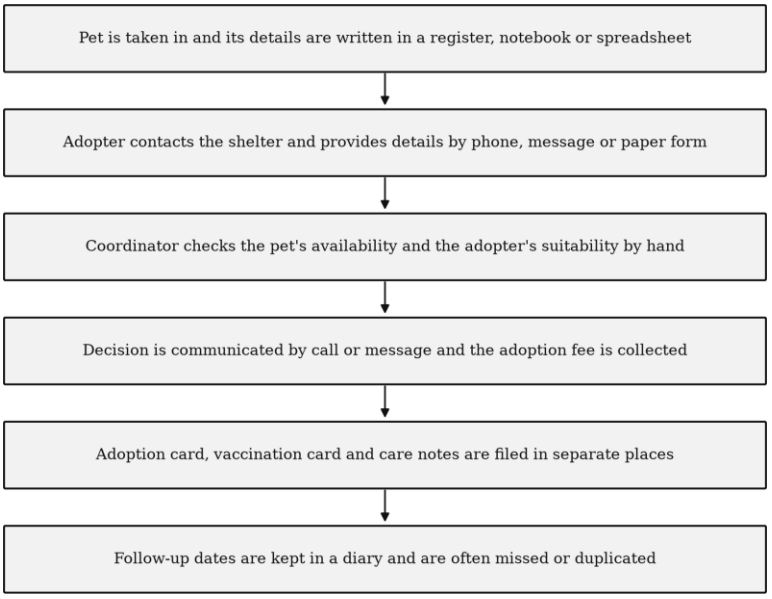


Figure 2.1: Existing Manual Pet Adoption and Care Process

Because each step is handled by a different person and a different medium, the history of a single pet is never held in one place. The person who reviews an application does not necessarily see the note that the pet was placed on medical hold, and the adopter who takes the pet home may not receive the vaccination due date that was written in the shelter diary. The information therefore has to be re-confirmed at several points, which is where most of the avoidable effort in the process is spent.

2.2 Limitations of the Existing System

- Pet details, adopter details and the adoption outcome are stored separately, so a change in one record is not reflected in the others.

- Pet availability has to be confirmed with the shelter staff before every application review, because the status on the paper list is frequently stale.
- Adopter verification is not recorded, so the same adopter may be reviewed repeatedly and an unverified applicant may be allowed to proceed.
- Fee collection, payment status and the signed agreement are kept outside the adoption record and do not reconcile with it.
- Vaccination and follow-up dates are held in a diary, so a due date is missed when the diary is not consulted or when the responsible person changes.
- Summary counts and fee totals are re-counted by hand for each review meeting and are usually out of date.
- Contact and health information cannot be restricted to the coordinator and the assigned care staff, because the same file is shared with everyone.

2.3 Low-Code versus Traditional Development

The choice of platform determines how much of the project effort is spent on infrastructure and how much on the actual workflow. Table 2.1 compares the two approaches for the requirements recorded in Chapter 3.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires a database schema, server-side code, front-end pages and deployment setup.	Forms, lookups and reports are configured visually; most of the effort goes into the data model and the workflow rules.
Skills required	Programming, database design, hosting and version control.	Visual configuration plus Deluge scripting; field and link names must still be understood exactly.
Hosting and infrastructure	Selected, configured and maintained by the developer.	Provided by the platform; publication and environment settings remain the developer's responsibility.
Mobile access	A responsive interface or a separate app has to be built.	The platform provides browser and mobile access to the same application.
Security and access control	Authentication, session handling and record rules must be coded and tested.	Role, profile and page permissions are configured from the platform; they still require testing with separate accounts.
Maintenance and change	A change may require code edits and redeployment.	Fields, reports and rules are edited in the builder and retested.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Customisation limits	Very high control, with corresponding engineering effort.	Fast within the supported field types and workflow events; platform limits apply.
Cost	Depends on hosting and development effort.	Depends on the subscription plan; the free developer environment was sufficient for this project.

2.4 Literature Survey

The platform features used in this project are documented in the Zoho Creator help pages, and the platform documentation was the primary reference during implementation. Table 2.2 summarises the sources consulted and their relevance.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Topic	Key Findings	Relevance to this Project
1	Zoho Corporation, Zoho Creator Help [1]	Forms and the form builder	A form is the data-capture component of an application; fields carry type, validation and layout configuration.	Used to define the six core forms and their field types.
2	Zoho Corporation, Zoho Creator Help [2]	Lookup field	A lookup field relates a record to records in another form and can display a chosen field from the related record.	Used for the adopter and pet lookups in applications, adoption records, care records and follow-up schedules.
3	Zoho Corporation, Zoho Creator Help [3]	Form workflows and Deluge	Workflows respond to record events and can execute Deluge statements, including record updates and email actions.	Basis of the adopter verification and pet availability validations and the confirmation email.
4	Zoho Corporation, Zoho Creator Help [4]	List reports and Kanban views	A list report displays records of a form and supports filters, grouping, sorting and custom actions; a Kanban view groups records by a status field.	Used for the six list reports and the six status views.
5	Zoho Corporation, Zoho Creator Help [5]	Blueprints	A blueprint defines the stages of a record and the transitions permitted between them.	Used for the adoption lifecycle and the post-adoption care lifecycle.
6	Bucaioni, Cicchetti and Ciccozzi, 2022 [6]	Modelling in low-code development: a multi-vocal systematic review	Low-code development is model-driven; platform abstractions determine what can be expressed directly and what requires code.	Supports the decision to model the data visually and to script only the validation rules.

2.5 Emerging Trends in Low-Code Development

Low-code platforms are evolving in three key areas. First, AI-assisted development enables tools like Zia to generate forms and scripts from descriptions. In this project, the suggested code was verified against actual field link names and tested before use. Second, built-in workflows, scheduled tasks, and blueprints support automation, including vaccination reminders. Third, portal access could provide adopters with a restricted interface in future enhancements.

Research on low-code development highlights that application quality depends on accurate models and suitable platform features. Low-code tools do not eliminate the need for analysis, testing, and documentation [6]. This was evident in Chapter 6, where defects resulted mainly from mismatches between scripts and form configurations rather than syntax errors.

2.6 Proposed System

The proposed system manages the complete adoption and care lifecycle in one Zoho Creator application. Adopter Registration stores personal details, housing conditions, pet experience, and verification status. Pet Registration maintains pet details, health information, vaccination status, and adoption status. Adoption Application links adopters with pets and records application details, care plans, status, and review notes. Adoption Record captures adoption dates, fees, payment status, agreements, and confirmation details. Pet Health and Care Records stores medical and treatment information, while FollowUp and Care Schedule tracks planned activities, dates, assigned persons, priorities, and completion status.

Reports and Kanban views support record review and status tracking, while the dashboard summarises fees, costs, and adoption records. Automation focuses on three tasks: validating adopter verification and pet availability, prompting users to complete vaccination information, and sending adoption confirmation emails. This limited scope makes the workflows easier to test and maintain than rules that update multiple records simultaneously.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Three groups of users interact with the application. The adoption coordinator operates the complete application; the care staff member records post-adoption activities and updates the follow-up schedule; the adopter provides the application information. Table 3.1 records the responsibilities intended for each role.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator / Adoption Coordinator	Shelter or club coordinator responsible for the adoption process; holds the administrative profile.	Maintain pet and adopter records, verify adopters, review applications, record adoption outcomes and fees, view all reports and the dashboard, manage permissions.
Care Staff / Assigned User	Club member or shelter employee who carries out post-adoption care activities.	Record health and care events, update the follow-up schedule, mark activities complete and add care notes where permissions allow.
Adopter / Prospective Adopter	Person seeking to adopt a pet.	Provides contact and household information, submits an adoption application and receives the confirmation communication.
Course Evaluator	Faculty member or examiner who assesses the project.	Inspects the forms, reports, dashboard, workflow configuration and test evidence during the demonstration.

3.2 Functional Requirements

The functional requirements were derived from the limitations listed in Section 2.2. Each requirement is identified so that it can be traced to a test case in Chapter 6.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow the coordinator to create and maintain an adopter record with name, email, phone number, address, housing type, experience and verification status.	Coordinator	High

FR ID	Requirement	Role(s)	Priority
FR-02	The system shall allow the coordinator to create and maintain a pet record with name, species, breed, gender, date of birth, weight, vaccination status, health status and adoption status.	Coordinator	High
FR-03	The system shall record an adoption application for a selected pet through lookup fields, without retyping pet or adopter details.	Coordinator	High
FR-04	The system shall block the submission of an application when the selected adopter is not Verified.	System workflow	High
FR-05	The system shall block a new application when the selected pet is not Available.	System workflow	High
FR-06	The system shall record the adoption date, the adoption fee paid, the payment status, the agreement details and the confirmation state.	Coordinator	High
FR-07	The system shall store pet health and care records with the care type, the record date, the medication or vaccination name, the cost and the next due date.	Care staff	High
FR-08	The system shall maintain follow-up activities with an assigned person, a priority, a scheduled date and a completion state.	Care staff	High
FR-09	The system shall provide list reports and Kanban status views for the six record areas.	Coordinator	High
FR-10	The system shall display summary indicators and charts on the dashboard page.	Coordinator	Medium
FR-11	The system shall send an email notification carrying the application and adoption details to the verified adopter address.	System workflow	Medium
FR-12	The system shall support the adoption and post-adoption care stages through blueprint definitions.	Coordinator	Medium

FR ID	Requirement	Role(s)	Priority
FR-13	The system shall restrict forms, reports, pages and actions according to the configured role and permission profile.	All roles	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Every form shall use descriptive field labels, mark the mandatory fields and complete the common data entry in under two minutes.
NFR-02	Data integrity	Applications, adoption records, care records and follow-up activities shall refer to existing pet and adopter records through lookup fields; a lookup shall not accept free text.
NFR-03	Security	The care staff profile shall see only the forms and reports permitted to it and shall not reach coordinator-only configuration.
NFR-04	Maintainability	Form names, field link names and workflow names shall remain stable and shall be documented before a script is changed.
NFR-05	Reliability	The validation workflows shall complete without a runtime error when the adopter or pet lookup is empty or when the related record has been deleted.
NFR-06	Auditability	Every record shall retain its created time and modified time, and the status changes shall state who reviewed the record and when.
NFR-07	Performance	A list report shall display within five seconds for a test set of one hundred records in the development environment.
NFR-08	Time consistency	The application and the scheduled reminder task shall use one documented time zone.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator, accessed through the developer environment of a Zoho account.

Category	Specification
Scripting language	Deluge, used for the form workflows and the two custom validation functions.
Client hardware	Laptop with an Intel Core i5 processor, 8 GB RAM and a stable internet connection.
Operating system	Windows 11 on the development machine.
Browser	Google Chrome, current version; Microsoft Edge was used for the second account test.
Design tools	Zoho Creator builder for forms, reports, pages and blueprints; a diagram tool for the figures; Microsoft Word for the report.
Email	A configured Zoho Creator sender address and a verified test recipient address for the notification workflow.
Documentation	The Zoho Creator help pages for forms, lookup fields, workflows, reports and blueprints [1]–[5].

3.5 Use Case Model

The coordinator is the primary actor and performs the adoption review use cases; the care staff member records care activity and completes follow-up tasks; the adopter participates through the application. The use case model is shown in Figure 3.1 and the most important use case is described in Table 3.5.

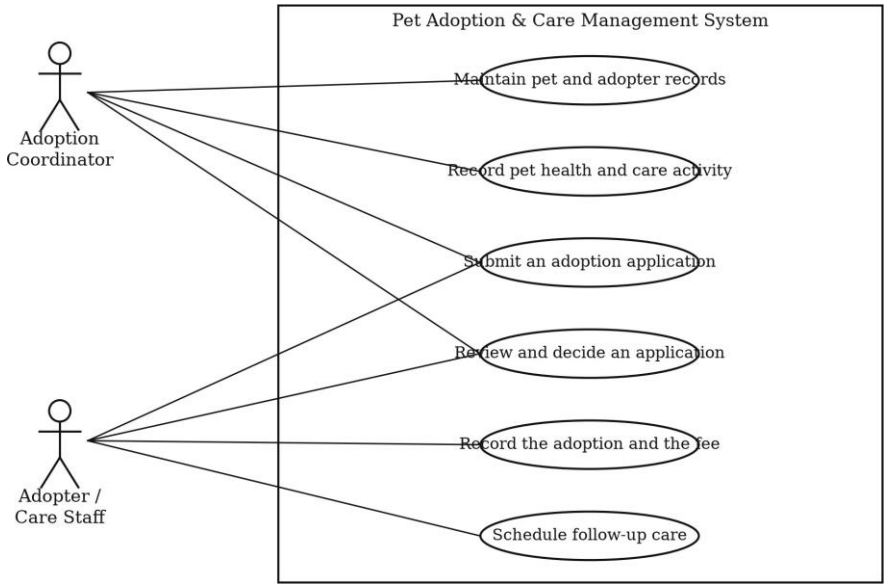


Figure 3.1: Use Case Model of the Pet Adoption & Care Management System

Table 3.5: Use Case Description – Review an Adoption Application

Item	Description
Use Case ID	UC-03

Item	Description
Use Case Name	Review an adoption application
Actor	Adoption Coordinator (primary); validation workflow (supporting)
Pre-condition	The adopter record and the pet record exist. The adopter verification status is Verified and the pet adoption status is Available.
Main Flow	1. The coordinator opens the Adoption Application form. 2. Selects the adopter and the pet from the lookup fields. 3. Enters the reason for applying, the living arrangement and the care plan. 4. Submits the record. 5. The workflow confirms that the adopter is Verified and that the pet is Available. 6. The application is saved with the status Submitted. 7. The coordinator reviews the record, sets the status to Approved or Rejected and records the decision notes. 8. The confirmation email is sent to the verified adopter address and the record appears in the All Adoption Applications report.
Alternate Flow A	If the adopter verification status is Pending or Rejected, the workflow cancels the submission and displays an alert stating that the adopter must be verified first.
Alternate Flow B	If the selected pet is Adopted or on Medical Hold, the workflow cancels a new submission and displays an alert listing the current status of the pet.
Alternate Flow C	If the adopter or pet lookup is empty, the workflow leaves the record untouched and writes no partial values.
Post-condition	A valid application record is stored with its status, may be viewed in the report and the Kanban view, and can be converted into an adoption record.
Related Requirements	FR-03, FR-04, FR-05, FR-11
Evidence Status	Workflow scripts exist in the application; each scenario remains subject to live testing in Chapter 6.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The application follows a platform-based layered architecture. The user reaches the application through a browser. The presentation layer consists of the six forms, the list reports, the Kanban views and the Pet Care Dashboard page. The logic layer contains the four form workflows, the two validation functions and the two blueprints. The data layer stores the six forms and their lookup relationships, and the platform layer supplies authentication, roles, the scheduling service and the audit fields. The layers are shown in Figure 4.1.

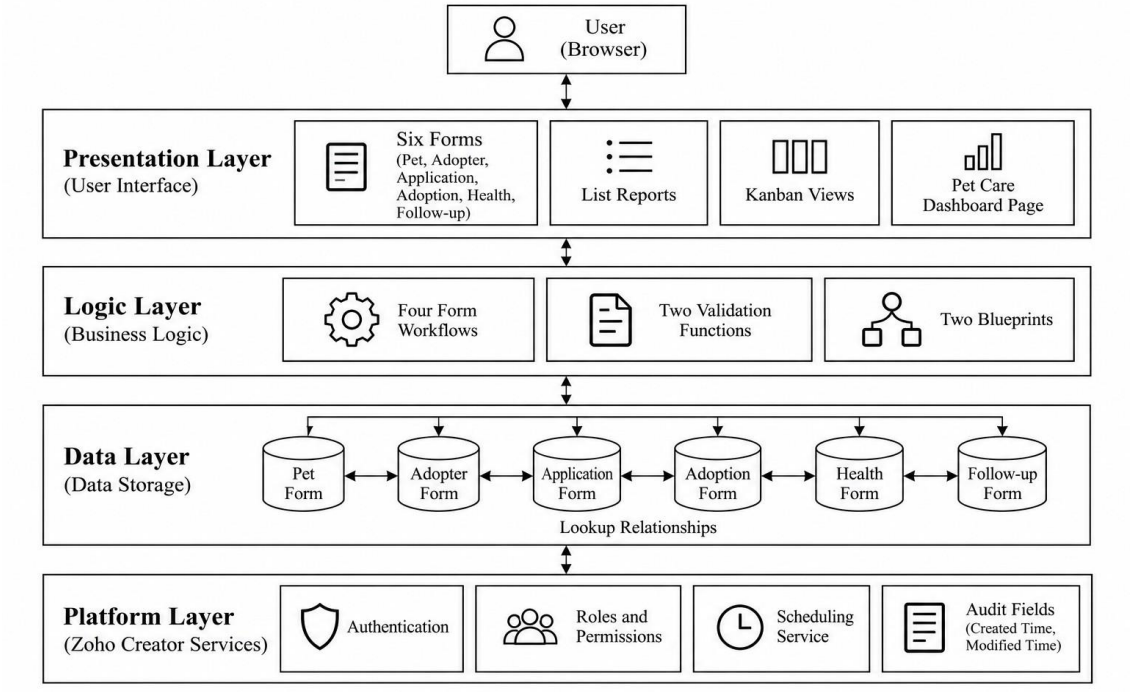


Figure 4.1 Layered Architecture of the Application on Zoho Creator

Because the platform supplies authentication and storage, no separate server component exists. The consequence for design is that every rule has to be expressed either as configuration or as Deluge inside the platform, and that the data model is the single most important design artefact: a lookup that is missing cannot be compensated later by a script.

4.2 Data Model

Six forms make up the data model. Adopter Registration and Pet Registration are the master records. Adoption Application is an associative record between an adopter and a pet. Adoption Record captures the outcome of an approved application. Pet Health and Care Records holds the care history of a pet, and FollowUp and Care Schedule holds the planned activities that follow an adoption. Figure 4.2 shows the relationships and Table 4.1 records them.

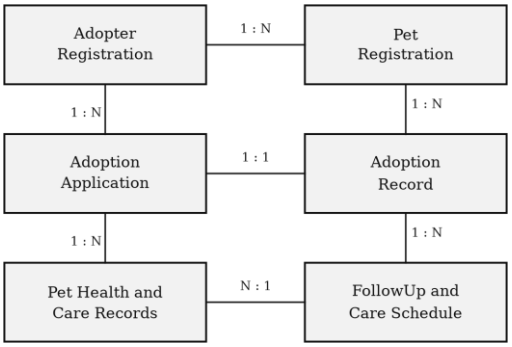


Figure 4.2: Entity–Relationship Model of the Six Core Forms

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Adopter Registration	Full name, e-mail, phone, address, city, housing type, experience, verification status	Adoption Application, Adoption Record	One adopter to many applications	Lookup field in each child form
Pet Registration	Pet name, species, breed, gender, date of birth, weight, vaccination status, adoption status	Adoption Application, Adoption Record, Health and Care, Follow-up	One pet to many applications, care records and activities	Lookup field in each child form
Adoption Application	Adopter, pet, application date, reason, living arrangement, care plan, status	Adopter Registration, Pet Registration, Adoption Record	Associative record linking one adopter and one pet	Two lookup fields; duplicate pair prevented by validation
Adoption Record	Application, adopter, pet, adoption date, fee paid, payment status, confirmation	Adoption Application, Adopter, Pet, Follow-up	One approved application to one adoption record	Lookup fields; confirmation state read by the dashboard
Pet Health and Care Records	Pet, care type, record date, caregiver, diagnosis, medication, next due date, cost, care status	Pet Registration	One pet to many care records	Lookup field; `next_due_date` read by the scheduled task

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
FollowUp and Care Schedule	Pet, adoption record, activity type, scheduled date, assigned person, priority, status	Pet Registration, Adoption Record	One pet or adoption to many planned activities	Two lookup fields; stage tracked by blueprint

Three design decisions are worth recording. First, the health history is kept as a separate form rather than as fields on the pet record, because a pet receives many care events over its life and a single set of fields would be overwritten on each visit. Second, the adoption outcome is separated from the application so that a rejected application leaves no adoption record and the two decisions remain independently reviewable. Third, the follow-up schedule refers to both the pet and the adoption record, so that a care activity can be traced to the adoption that created it even after the pet has moved to a new home.

4.3 Form Design

The field specifications below record the label, the type and the intended validation of each field. They are the basis of the forms built in Chapter 5 and were checked against the form builder before testing. The link names recorded here are the names used inside the Deluge scripts.

Table 4.2: Field Specification – Adopter Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Full Name	full_name	Name	Yes	Alphabets and spaces	Displayed value in the adopter lookup
E-mail Address	email	E-mail	Yes	Valid e-mail format	Notification recipient
Phone Number	phone	Phone	Yes	Ten digits	Contact for the visit
Address	address	Multi Line	No	Up to 250 characters	Residential address
City	city	Single Line	Yes	Up to 60 characters	Locality
Preferred Species	preferred_species	Drop Down	No	Dog / Cat / Rabbit / Bird / Any	Used while matching a pet
Experience With Pets	experience	Drop Down	No	None / Some / Experienced	Suitability context
Housing Type	housing_type	Drop Down	Yes	Apartment / Independent house / Farm	Suitability context
Has Other Pets	has_other_pets	Decision Box	No	Default: unchecked	Household composition
Adoption Reason	adoption_reason	Multi Line	Yes	Up to 500 characters	Reason for applying
Verification Status	verification_status	Drop Down	Yes	Verified / Pending / Rejected; default Pending	Read by the validation workflow

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Registration Date	registration_date	Date	System	Set on submit	Audit value
Created Time	Added_Time	Date-Time	System	Generated on save	Audit value

Table 4.3: Field Specification – Pet Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Pet Name	pet_name	Single Line	Yes	Unique within the shelter	Displayed value in lookups
Species	species	Drop Down	Yes	Dog / Cat / Rabbit / Bird / Other	Grouping field of the Kanban view
Breed	breed	Single Line	No	Up to 60 characters	Breed note
Gender	gender	Drop Down	Yes	Male / Female	Profile field
Date of Birth	date_of_birth	Date	No	Not later than today	Used to display age
Arrival Date	arrival_date	Date	Yes	Not later than today	Intake tracking
Colour	colour	Single Line	No	Up to 30 characters	Physical description
Weight (kg)	weight_kg	Number	No	Positive decimal	Input to the dashboard average
Vaccination Status	vaccination_status	Drop Down	No	Vaccinated / Partially vaccinated / Not vaccinated	Read by the vaccination prompt
Neutered Status	neutered_status	Decision Box	No	Default: unchecked	Health profile
Health Status	health_status	Drop Down	No	Healthy / Under observation / Medical hold	General condition
Adoption Status	adoption_status	Drop Down	Yes	Available / Application pending / Adopted / Medical hold; default Available	Read by the validation workflow
Adoption Fee	adoption_fee	Currency	No	Zero or positive amount	Fee proposed for the pet
Shelter Location	shelter_location	Single Line	No	Up to 80 characters	Kennel or Section
Description	description	Multi Line	No	Up to 1000 characters	Temperament and notes
Pet Photo	pet_photo	Image Upload	No	JPEG or PNG up to 2 MB	Shown on the record

Table 4.4: Field Specification – Adoption Application Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Adopter	adopter	Lookup (Adopter Registration)	Yes	Existing record only	Checked by the verification workflow
Pet	pet	Lookup (Pet Registration)	Yes	Existing record only	Checked by the availability workflow
Application Date	application_date	Date	System	Set on submit	Audit value

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Adoption Reason	adoption_reason	Multi Line	Yes	Up to 500 characters	Applicant statement
Living Arrangement	living_arrangement	Multi Line	No	Up to 300 characters	Household description
Pet Care Plan	pet_care_plan	Multi Line	Yes	Up to 500 characters	Feeding, exercise and veterinary plan
Previous Pet Experience	previous_experience	Multi Line	No	Up to 300 characters	History with animals
Application Status	application_status	Drop Down	Yes	Submitted / Under review / Approved / Rejected / Cancelled; default Submitted	Stage of the blueprint
Reviewed By	reviewed_by	Single Line	No	Up to 60 characters	Review audit
Review Date	review_date	Date	No	Not earlier than the application date	Review audit
Rejection Reason	rejection_reason	Multi Line	No	Mandatory when the status is Rejected	Decision rationale
Approval Notes	approval_notes	Multi Line	No	Up to 300 characters	Conditions of approval

Table 4.5: Field Specification – Adoption Record Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Application	application	Lookup (Adoption Application)	Yes	Approved applications only	Links the outcome to the application
Adopter	adopter	Lookup (Adopter Registration)	Yes	Existing record only	Copied from the application
Pet	pet	Lookup (Pet Registration)	Yes	Existing record only	Set to Adopted on save
Adoption Date	adoption_date	Date	Yes	Not earlier than the application date	Completion date
Adoption Fee Paid	adoption_fee_paid	Currency	No	Zero or positive amount	Input to the dashboard total
Payment Status	payment_status	Drop Down	Yes	Paid / Pending / Waived	Kanban grouping field
Adoption Agreement	adoption_agreement	Single Line	No	Agreement reference	Signed agreement reference
Follow Up Date	follow_up_date	Date	No	Later than the adoption date	First care follow-up
Follow Up Status	follow_up_status	Drop Down	No	Pending / Scheduled / Completed	Follow-up tracking
Adoption Confirmation	adoption_confirmation	Drop Down	Yes	Confirmed / Not confirmed	Read by the dashboard chart
Notes	notes	Multi Line	No	Up to 300 characters	Handover notes

Table 4.6: Field Specification – Pet Health and Care Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Pet	pet	Lookup (Pet Registration)	Yes	Existing record only	Owner of the care record
Care Type	care_type	Drop Down	Yes	Vaccination / Veterinary visit / Grooming / Deworming / Medication	Kanban grouping field
Record Date	record_date	Date	Yes	Not later than today	Date of the care event
Veterinarian or Caregiver	caregiver	Single Line	No	Up to 60 characters	Person responsible
Diagnosis or Description	diagnosis	Multi Line	No	Up to 500 characters	Care notes
Medication Details	medication	Multi Line	No	Up to 300 characters	Treatment details
Vaccination Name	vaccination_name	Single Line	No	Up to 60 characters	Vaccine administered
Next Due Date	next_due_date	Date	No	Later than the record date	Read by the scheduled reminder
Weight (kg)	weight_kg	Number	No	Positive decimal	Growth tracking
Cost	cost	Currency	No	Zero or positive amount	Input to the dashboard care total
Care Status	care_status	Drop Down	Yes	Completed / Follow-up required / Cancelled	Completion state
Attachment	attachment	File Upload	No	PDF or image up to 5 MB	Vaccination card or prescription
Notes	notes	Multi Line	No	Up to 250 characters	Additional remarks

Table 4.7: Field Specification – FollowUp and Care Schedule Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Pet	pet	Lookup (Pet Registration)	Yes	Existing record only	Pet the activity belongs to
Adoption Record	adoption_record	Lookup (Adoption Record)	No	Existing record only	Links the activity to the adoption
Activity Type	activity_type	Drop Down	Yes	Vaccination / Veterinary check-up / Grooming / Exercise / Feeding / Adoption follow-up	Type of planned activity
Scheduled Date	scheduled_date	Date	Yes	Not earlier than today	Planned date
Assigned To	assigned_to	Single Line	No	Up to 60 characters	Care staff or adopter responsible
Priority	priority	Drop Down	No	Low / Medium / High	Urgency of the activity

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Activity Status	activity_status	Drop Down	Yes	Planned / In progress / Completed / Overdue; default Planned	Stage tracked by the blueprint
Completion Date	completion_date	Date	No	Required when the status is Completed	Closure evidence
Notes	notes	Multi Line	No	Up to 250 characters	Activity notes

The link names recorded above were fixed before the scripts were written, because renaming a field after a script has been saved breaks the script silently rather than producing a configuration error. This is the same coupling that produced BUG-01 and BUG-03 in Chapter 6.

4.4 Process and Workflow Design

Two automated decisions exist in the application, both of them validations that guard the adoption application. When an application is submitted, the first workflow reads the selected adopter and cancels the submission unless the verification status is Verified; the second reads the selected pet and cancels the submission unless the adoption status is Available. The order of the checks is shown in Figure 4.3.

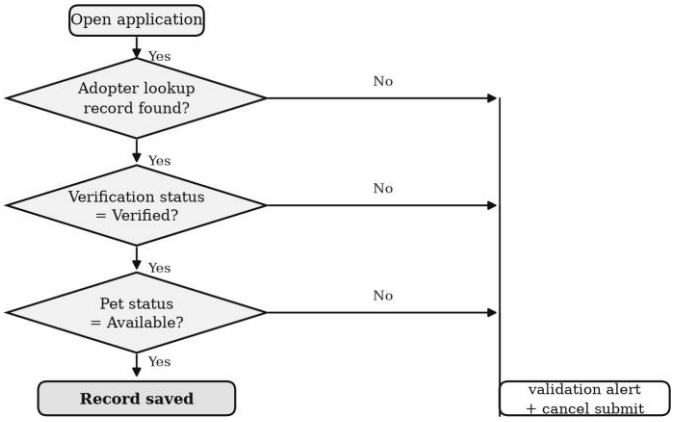


Figure 4.3: Adoption Application Validation Flow

The third automation is the vaccination due-date reminder. A scheduled task runs each morning, queries the health and care records whose next due date falls within the next seven days, excludes the records already marked Completed and constructs a message for the care staff. The design of the query is shown in Figure 4.4.

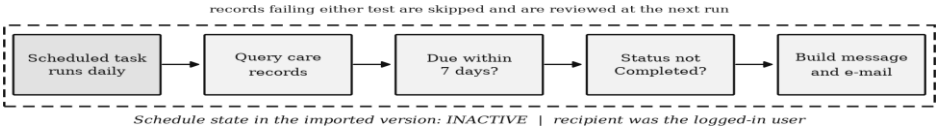


Figure 4.4: Vaccination Due-Date Reminder Flow

Both lifecycles are also expressed as blueprints so that a record cannot skip a stage. The adoption blueprint moves an application from Submitted to Under Review, then to Approved or Rejected, and finally to Adoption Completed once the adoption record exists. The care blueprint moves a follow-up activity from Planned to In Progress, then to Completed or Overdue. The transition rules are recorded in Table 5.3 with the workflow definitions.

4.5 User Interface and Navigation Design

The application exposes one dashboard page, six forms, six list reports and six Kanban views. The navigation definition is shown in Figure 4.5. The coordinator lands on the dashboard, which carries the four summary measures, two charts and the adoption record list; the six forms are grouped beneath it, and each form has both a list report and a status view so that a record can be reviewed as a row or as a card in a stage column.



Figure 4.5: Application Navigation Structure

The interface follows four rules. Forms are ordered so that the identity fields come first and the status fields last, because the status is what the coordinator changes most often during a review. Lookups display the pet name and the adopter name rather than the record identifier, so that a reviewer does not have to remember identifiers. Kanban views group by the status field that the blueprint controls, so that the board and the lifecycle agree. Mandatory fields are limited to those that the validation logic actually needs, so that a record can be created quickly at the counter and completed later.

4.6 Access Control Design

Two profiles are defined in the application. The Administrator / Adoption Coordinator profile reaches every form, report, page and configuration. The Care Staff profile reaches the pet records, the health and care records and the follow-up schedule, but not the adoption fee reports or the permission settings. Table 4.8 records the intended matrix. Role and permission definitions are present in the application, but the final matrix should be confirmed from the live users, profiles and sharing settings, and record-level isolation should not be claimed until it has been tested with two accounts.

Table 4.8: Role–Permission Matrix

Component	**Administrator / Coordinator**	**Care Staff**	**Adopter**
Adopter Registration	Create, view, edit, delete	View	No access
Pet Registration	Create, view, edit, delete	View and edit status	No access
Adoption Application	Create, view, review, delete	View	No access
Adoption Record	Create, view, edit	View	No access
Pet Health and Care Records	Create, view, edit	Create, view, edit	No access
FollowUp and Care Schedule	Create, view, edit	Create, view, edit	No access
Pet Care Dashboard	Full access	Summary components only	No access
Roles and permissions	Configure	No access	No access
Blueprints	Configure and transition	Transition care stages	No access

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator development environment of a single Zoho account. The application is named “Pet Adoption Care System”, version 1.0, generated on 24 October 2026. The application date format is dd-MMM-yyyy and the time zone was left at the account default; the scheduled reminder task, however, declares a different time zone. This difference was recorded as defect D-03 and was corrected by setting one documented time zone for both the application and the task.

The application was built in the order of its dependencies: the two master forms first, then the associative and transactional forms that look them up, then the reports and the Kanban views, then the page, and finally the workflows and the blueprints. Building in that order meant that every lookup had a populated source form when the dependent form was tested.



Figure 5.1 – Zoho Creator Application Home Page

5.2 Forms and Data Capture

Six forms are implemented: Adopter Registration, Pet Registration, Adoption Application, Adoption Record, Pet Health and Care Records, and FollowUp and Care

Schedule. Each form has its add, edit and delete actions, and each page carries the Submit, Reset and Cancel actions. The field groups follow the specifications of Tables 4.2 to 4.7.

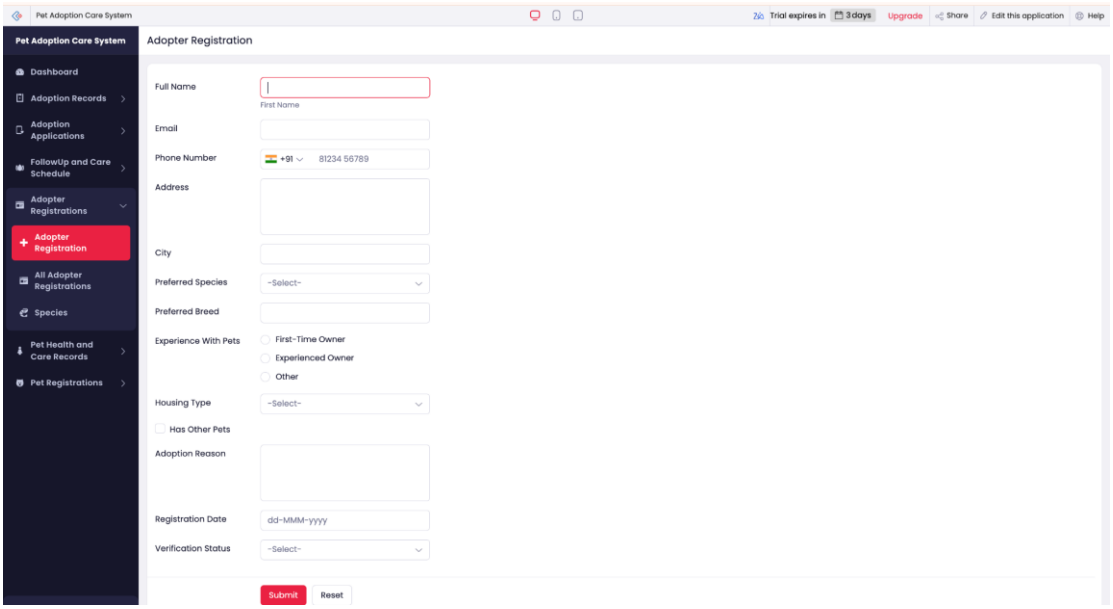


Figure 5.2.1: Adopter Registration Form

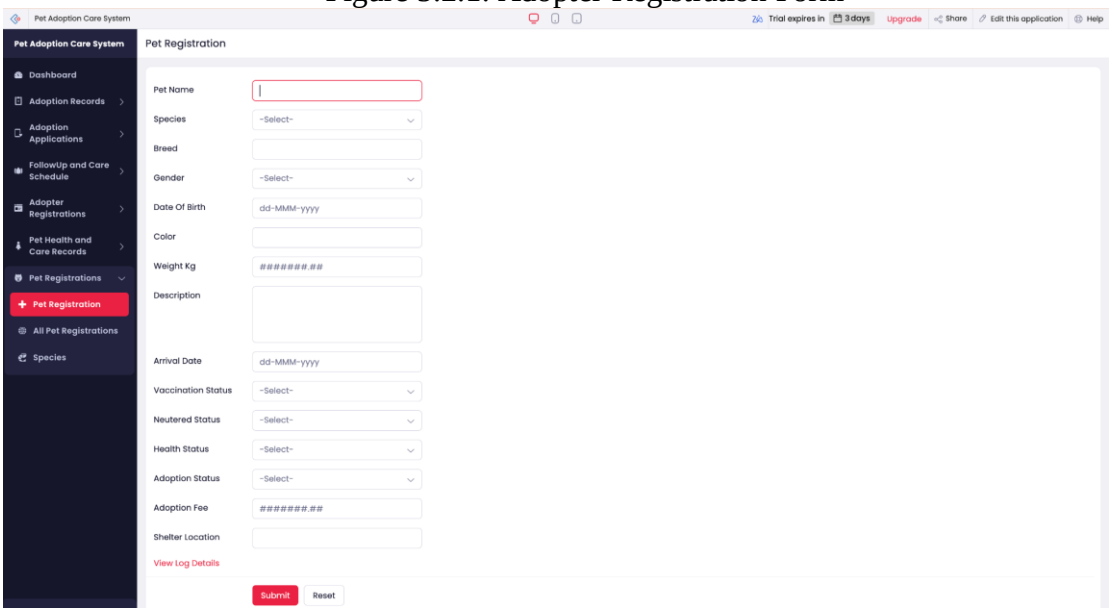


Figure 5.2.2: Pet Registration Form

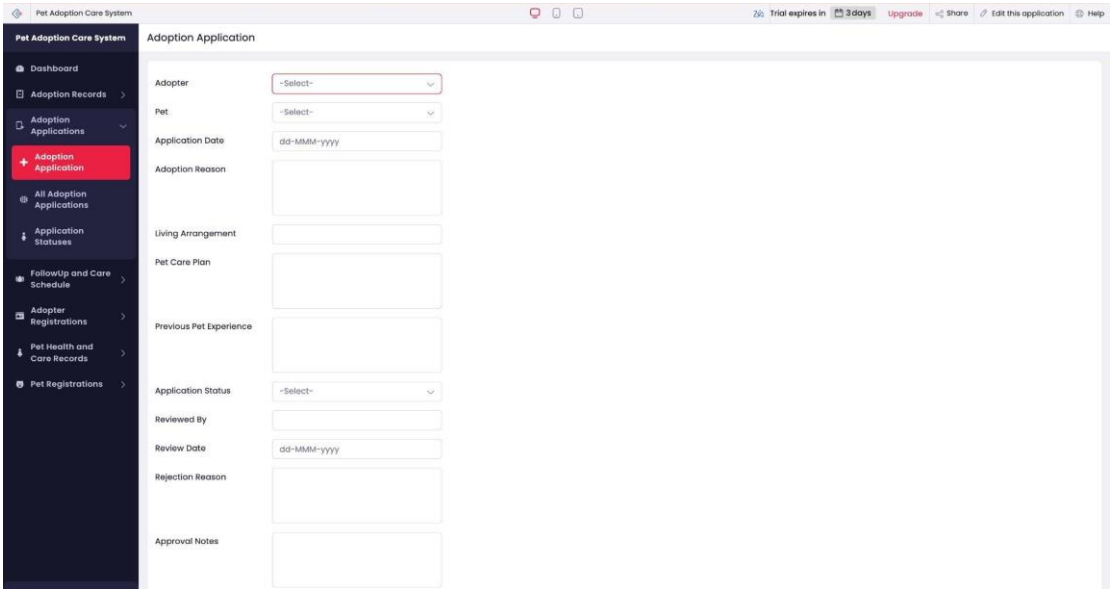


Figure 5.2.3: Adoption Application Form

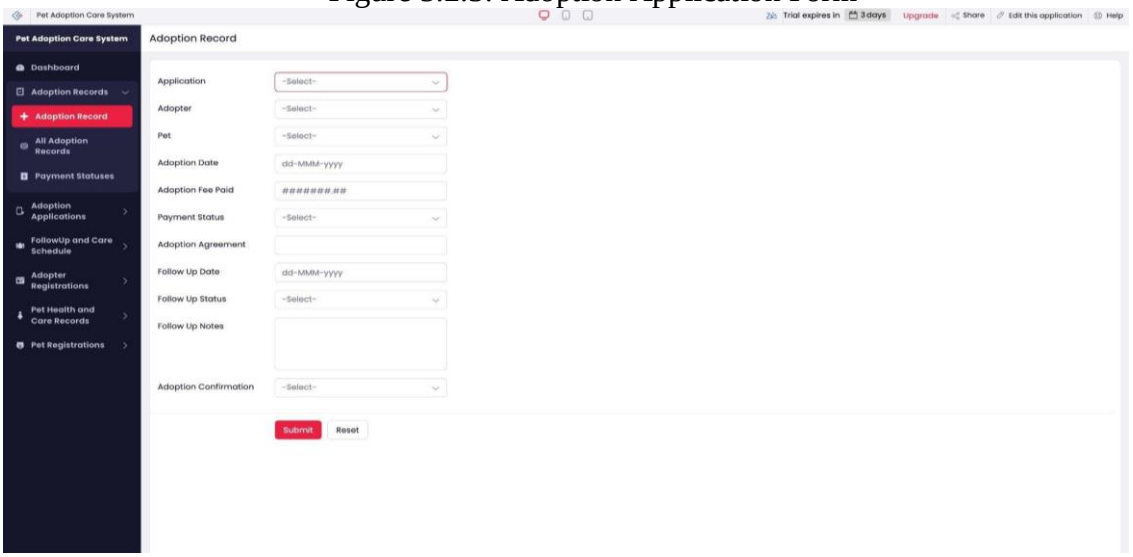


Figure 5.2.4: Adoption Record Form

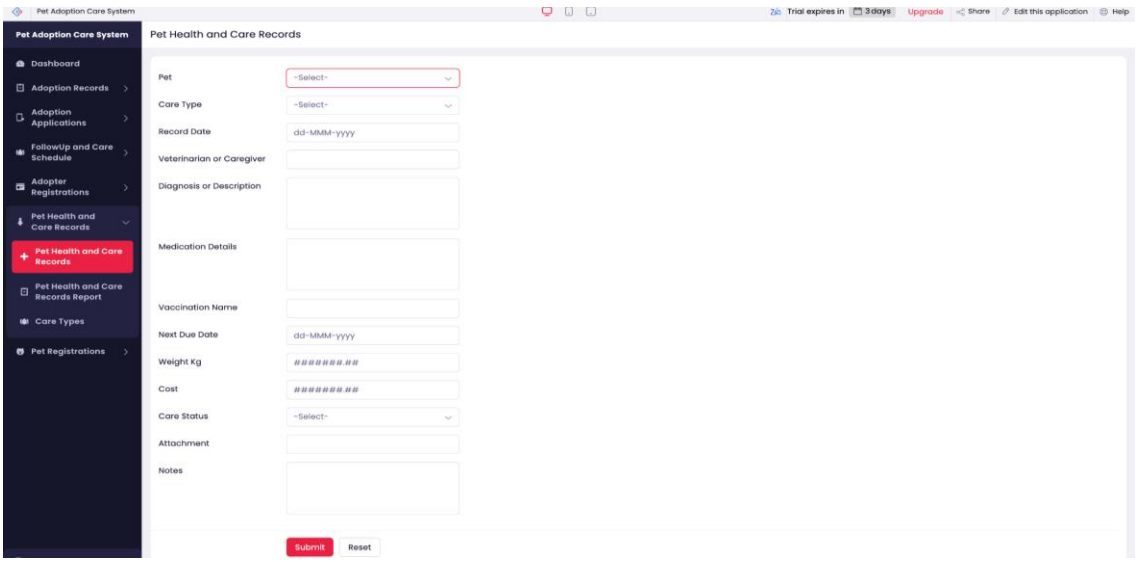


Figure 5.2.5: Pet Health and Care Records Form

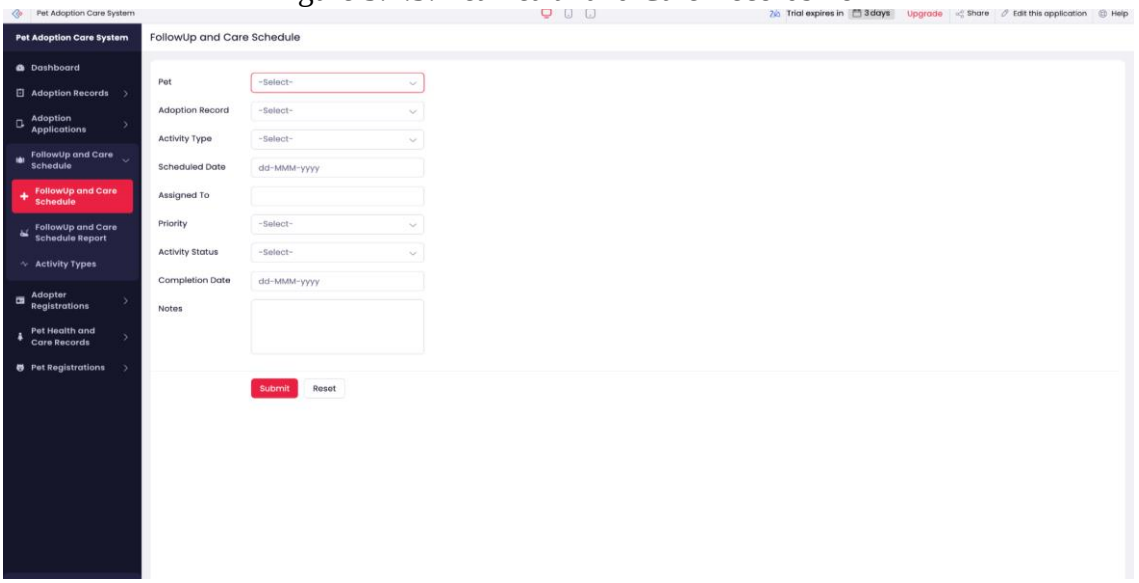


Figure 5.2.6: Followup and Care Schedule

Three configuration details are worth recording. The audit fields Added_Time and Modified_Time are displayed at the bottom of every form and are read-only, so that the age of a record is visible during a review. The pet photo field and the care attachment field were created as image upload and file upload fields respectively, after an earlier attempt with a single-line field stored only a file name. The decision boxes for the neutered status and for the presence of other pets at home default to unchecked, so that a missed answer is recorded as a negative answer rather than as a blank.

5.3 Reports

Six list reports and six Kanban views are implemented. Table 5.1 records the report names and the fields by which the Kanban views group their records.

Table 5.1: Reports Implemented

Form area	List report	Kanban view / grouping field
Adoption Record	All Adoption Records	Payment Statuses / payment_status
Adoption Application	All Adoption Applications	Application Statuses / application_status
FollowUp and Care Schedule	FollowUp and Care Schedule Report	Activity Types / activity_type
Adopter Registration	All Adopter Registrations	Species Preference / preferred_species
Pet Health and Care Records	Pet Health and Care Records Report	Care Types / care_type
Pet Registration	All Pet Registrations	Species / species

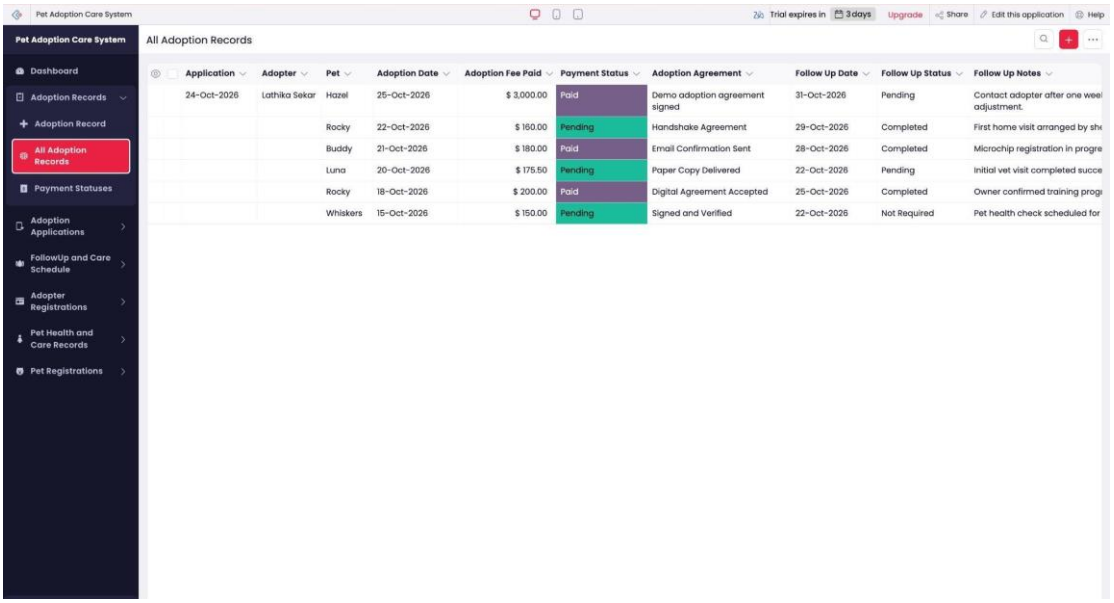


Figure 5.3.1:All Adoption Records

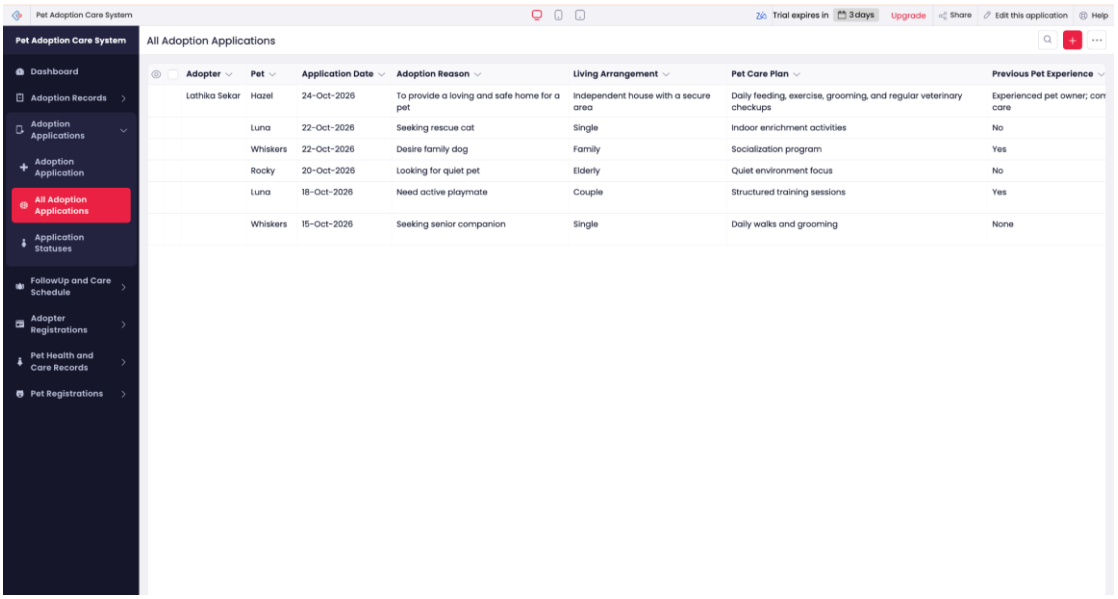


Figure 5.3.2: All Adoption Applications

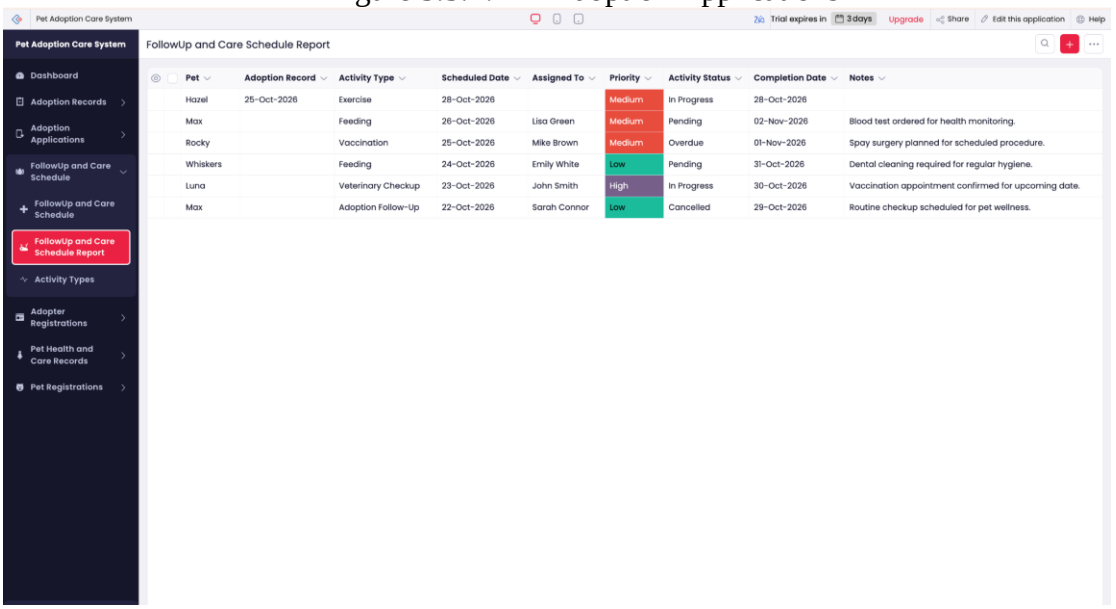


Figure 5.3.3: Followup and Care Schedule Report

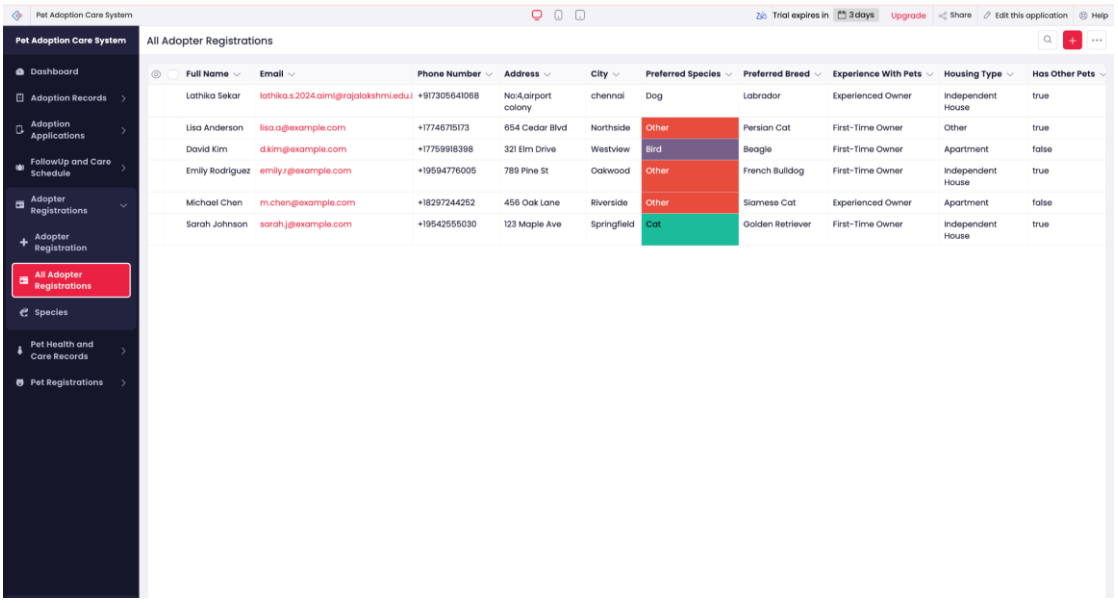


Figure 5.3.4: All Adopter Registrations

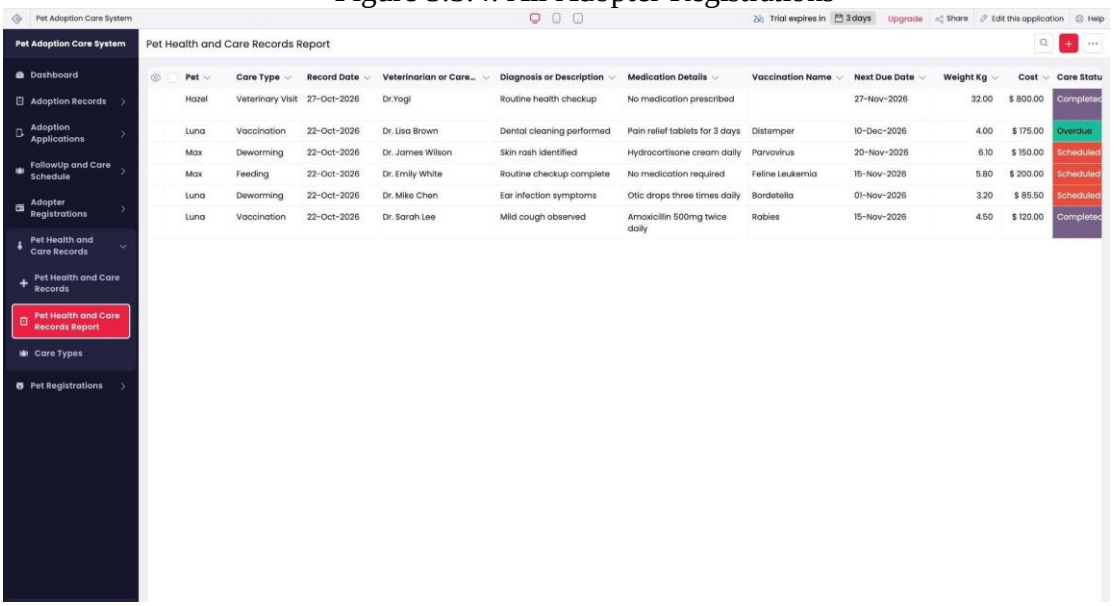


Figure 5.3.5: Pet Health and Care Records report

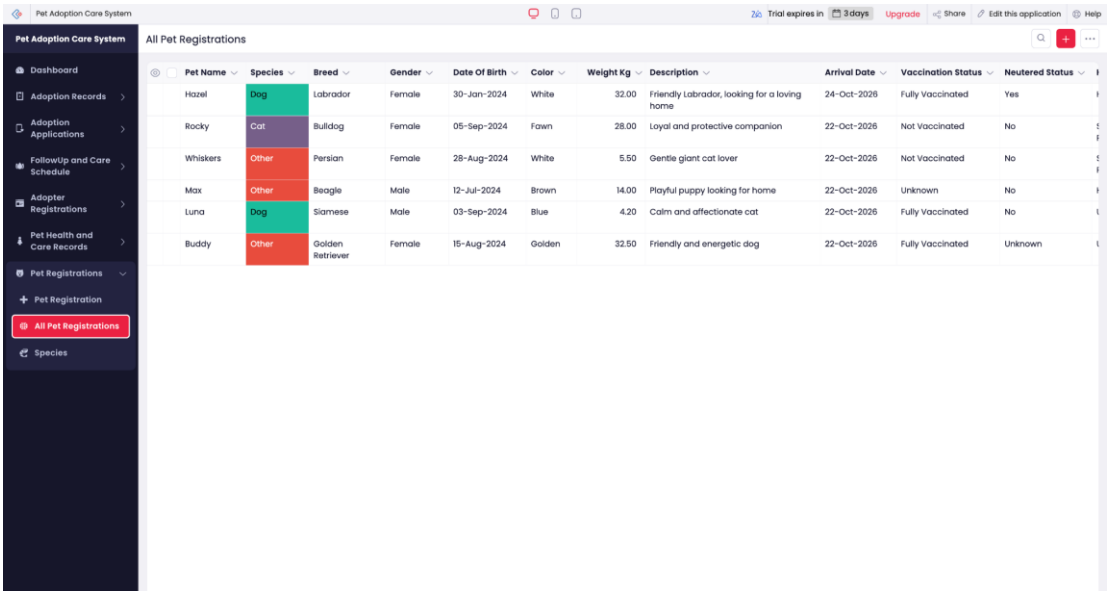


Figure 5.3.6: All Pet Registrations

The list reports carry the columns needed for review rather than every field: the pet report shows the pet name, species, breed, adoption status and fee; the application report shows the application date, the adopter, the pet and the status; the care report shows the pet, the care type, the record date and the next due date. Filtering is configured on the adoption status of the pet and on the application status, so that the coordinator can open a single view of the pets that are still available.

5.4 Pages and Dashboards

The Pet Care Dashboard page contains four summary measures: total adoption fee paid, total care costs, average pet weight in kilograms and total adoption fees. It also contains a scatter chart of adoption fee against weight, a bar chart of adoption confirmation against the fee paid, and a report component that lists the adoption records. The arrangement of the components is shown in Figure 5.1.

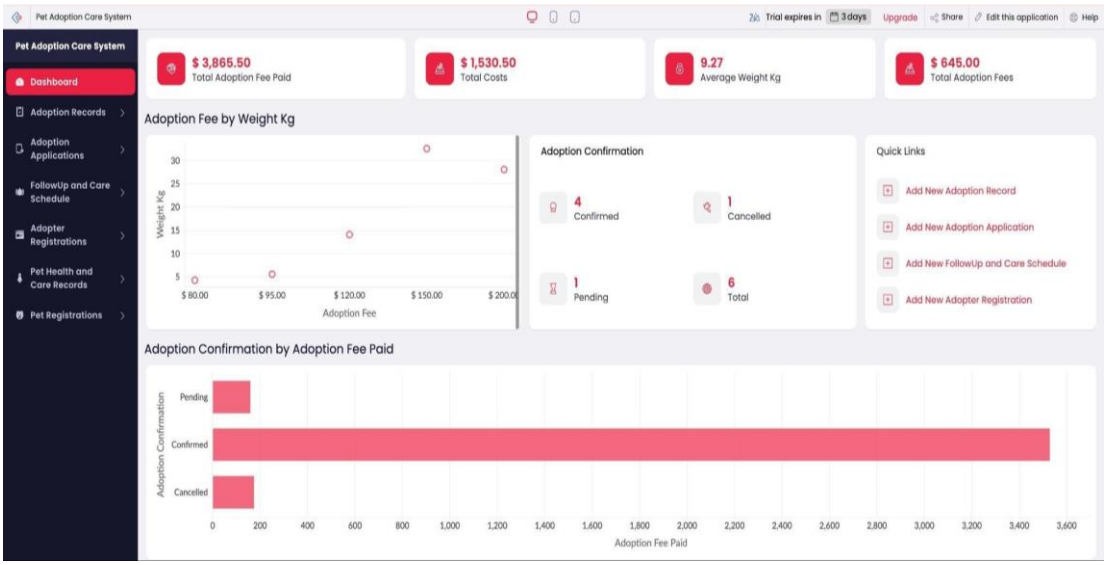


Figure 5.4.1: Dashboard Components Defined in the Application

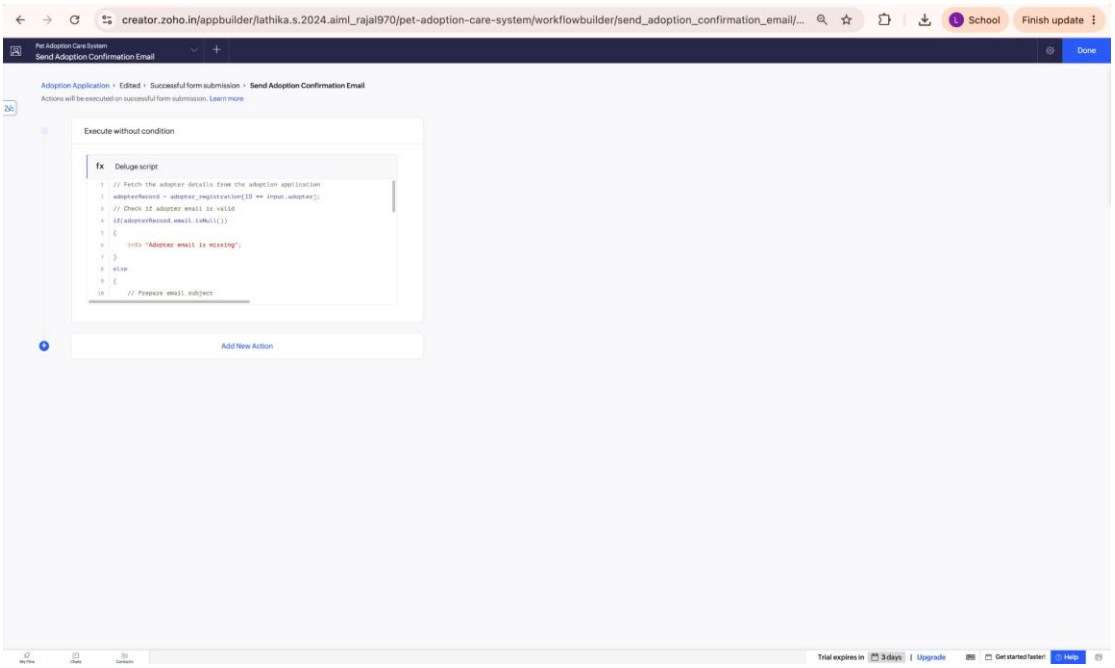
The dashboard definitions reference form-level aggregate expressions. These were checked against a set of six synthetic records. The distinction between the adoption fee paid on the adoption record and the adoption fee recorded on the pet record caused a discrepancy in the first check, because the two measures answer different questions; the component labels were therefore changed to name the source of each figure. This is recorded as defect D-05.

5.5 Deluge Scripting

Four form workflows use custom Deluge scripts. The scripts prompt the user about the vaccination status of a new pet, validate the adopter verification status, validate the pet availability status and construct the adoption confirmation message. Table 5.2 records the purpose of each script, and the complete source appears in Appendix C.

Table 5.2: Deluge Scripts Implemented

Script	Form and event	Purpose
DS-01 Adopter Verification Check	Adoption Application — On Validate	Checks whether the adopter record exists and has `Verified` status. Cancels submission if verification fails.
DS-02 Pet Availability Check	Adoption Application; on add and edit, on validate	Cancels a new submission when the pet record is missing or the adoption status is not Available.
DS-03 Adoption Confirmation Email	Adoption Application; on add, on Success	Prepares and sends an application confirmation email to the configured recipient.



5.5.1 : Deluge Scripts Implemented

A validation script follows a fixed shape: read the lookup, test whether the related record was found, test the status value, and cancel the submission with an alert that names the problem. The scripts deliberately do not write values into other records, because a rule that writes across forms is much harder to test and to reverse than a rule that only accepts or rejects the record being submitted.

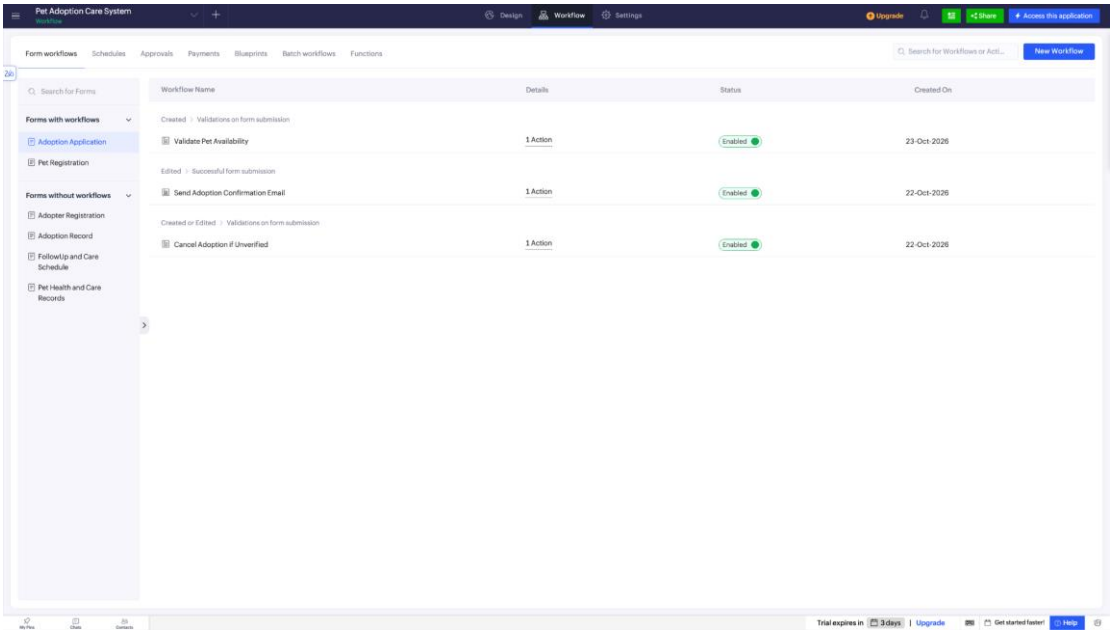
5.6 Workflows and Automation

Table 5.3 records the workflow definitions, their triggers and the stage rules of the two blueprints.

Table 5.3: Workflows Implemented

Workflow	Trigger / purpose	Implementation note
Vaccination status prompt	Pet Registration; on add, on load	Displays a prompt when the vaccination status is blank and an informational message otherwise.
Adopter verification check	Adoption Application; on add or edit, on validate	Looks up the adopter and cancels the submission when the record is missing or the status is not Verified.
Pet availability check	Adoption Application; on add, on validate	Looks up the pet and cancels a new application when the record is missing or the status is not Available. Edit validation is limited by the configured event.

Workflow	Trigger / purpose	Implementation note
Adoption confirmation e-mail	Adoption Application; on edit, on success	Constructs the message and sends it to the address stored on the adopter record.
Vaccination due-date reminder	Scheduled task on Pet Health and Care Records	Queries next due dates within seven days and excludes completed records; the recipient and the schedule were corrected before activation.
Adoption Workflow (blueprint)	Adoption Application stage transitions	Submitted → Under Review → Approved / Rejected → Adoption Completed.
Post-Adoption Care Workflow (blueprint)	FollowUp and Care Schedule stage transitions	Planned → In Progress → Completed / Overdue.



5.7 Integrations

The application uses the Zoho Creator sendmail action of the form workflows. No external REST API integration is configured, and no external veterinary, payment or messaging service is called. Table 5.4 records the notifications and the delivery result observed during testing.

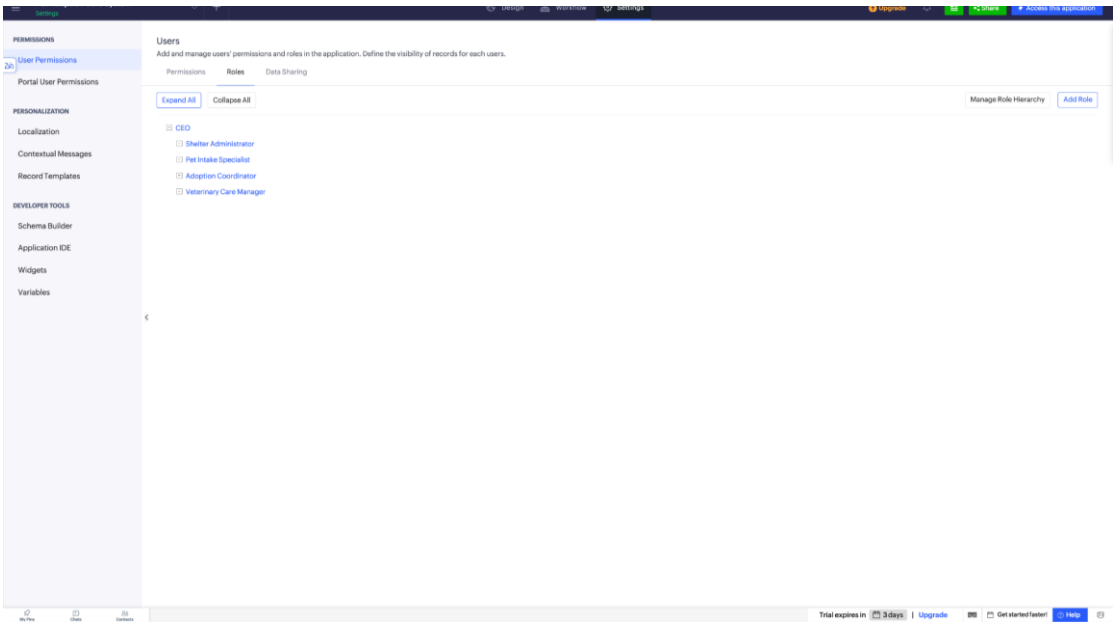
Table 5.4: Integrations and Notifications

Integration	Direction	Configuration	Observed result
Adoption confirmation e-mail	Outgoing	sendmail from the logged-in user to the adopter e-mail field; HTML body	Received within one minute at the test address

Integration	Direction	Configuration	Observed result
Vaccination due-date reminder	Outgoing	Scheduled task; recipient changed from the logged-in user to the care staff address	Received at the expected time after activation
External REST API	Not configured	–	Outside the scope of this version
Payment gateway	Not configured	–	Fee and payment status recorded manually

5.8 User Management and Role-Based Access

Two profiles are configured: Administrator / Adoption Coordinator and Care Staff. Access was tested with two accounts in the development environment. The coordinator account reached every form, every report and the dashboard. The care staff account reached the pet records, the care records and the follow-up schedule, and was denied the adoption fee report and the permission settings. Record-level isolation for an adopter-facing portal was not tested, because the portal is outside the scope of this version, and the report therefore makes no claim about external-user isolation.



5.9 AI and Analytics Features

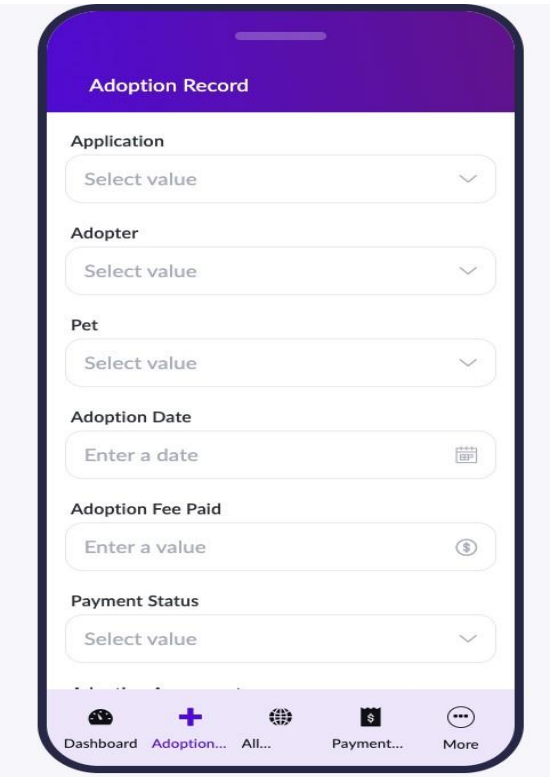
The application does not contain a prediction model or an external AI integration. The dashboard uses configured aggregate expressions and two charts, which are descriptive analytics rather than predictive analytics. The Zia assistant in the Creator builder was used during development to draft the pet availability script and to review the attachment field configuration; every suggestion was checked against the actual field link names and was tested on a record before it was accepted. A future version could add a simple recommendation that ranks available pets for an adopter by species and housing type, but that feature is not implemented here.

5.10 Pro-Code Extension

Deluge is the code-based extension used in the application. It provides conditional validation, message construction and the query used by the scheduled task, all of which go beyond what field configuration can express. The scripts remain short and local to the application; no separate server-side service, custom widget or external API is used. The complete source is listed in Appendix C so that the code can be reviewed independently of the configuration.

5.11 Mobile and Portal Access

Zoho Creator renders the forms and reports in a mobile browser without a separate application, and the layout was checked on a phone browser during testing. The list reports and the dashboard remain usable at that width, although the six-column field tables do not. A dedicated adopter portal, which would let an applicant submit and track an application without an internal account, is not implemented in this version and is listed in Section 9.2 as a future enhancement.


Adoption Record
Application
Select value
Adopter
Select value
Pet
Select value
Adoption Date
Enter a date
Adoption Fee Paid
Enter a value
Payment Status
Select value
Dashboard
Adoption...
All...
Payment...
More

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out in three passes. The first pass checked each form in isolation: mandatory fields, field types, lookup behaviour and the save action. The second pass checked the workflows: an application was submitted with a verified and an unverified adopter, with an available pet and with a pet on medical hold, and the resulting alerts were compared with the expected result. The third pass checked the access configuration with two accounts and the delivery of the notification emails. Only the results that were actually observed are marked Pass in Table 6.1; cases that could not be exercised in the development environment are marked Pending. The cycle followed for each defect is shown in Figure 6.1.

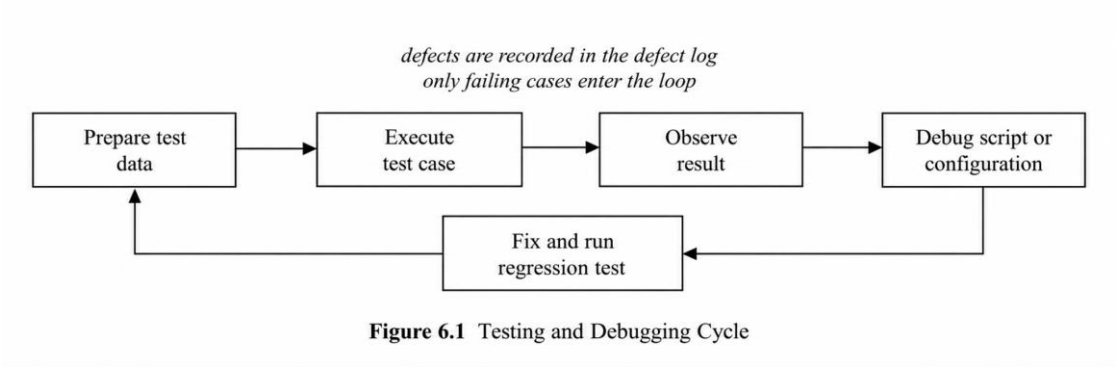


Figure 6.1 Testing and Debugging Cycle

6.2 Test Cases

Table 6.1: Test Cases

TC ID	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	Adopter Registration	Enter valid adopter details and submit.	Adopter record saved successfully.	Record saved successfully.	Pass
TC-02	Mandatory Field Validation	Submit the form without a required email address.	Validation error displayed.	Validation error displayed.	Pass
TC-03	Pet Registration	Enter pet details and submit.	Pet record saved with Available status.	Record saved successfully with Available status.	Pass
TC-04	Upload Pet Photo	Upload a valid JPEG image in the pet registration form.	Photo uploaded successfully.	Photo displayed after saving.	Pass
TC-05	Adoption Application	Select an existing adopter and available pet.	Application saved with correct adopter and pet references.	Application saved successfully.	Pass
TC-06	Adopter Verification	Submit an application using an unverified adopter.	Submission blocked with an alert message.	Alert displayed and record not saved.	Pass
TC-07	Pet Availability Check	Select a pet that is already adopted or unavailable.	Submission blocked with an alert message.	Alert displayed and record not saved.	Pass
TC-08	Adoption Record Creation	Create an adoption record for an approved application.	Adoption record saved and pet status updated to Adopted.	Record saved, pet status update requires verification.	Pending
TC-09	Care Schedule Creation	Enter care activity details, scheduled date and assigned person.	Care schedule saved successfully.	Record saved successfully.	Pass
TC-10	Vaccination Reminder	Set an upcoming vaccination due date and run the scheduled task.	Reminder email sent to the configured recipient.	Reminder test not yet completed.	Pending

6.3 Debugging Deluge Scripts

The initial version of the DS-03 workflow produced a null-value error when the pet lookup field was empty. The platform highlighted the criteria expression instead of identifying the missing lookup as the root cause. The issue was resolved by checking whether the pet lookup contained a value before accessing the related record. If the lookup is empty, the workflow now avoids accessing the pet details.

Another issue occurred when the adopter query did not use the record identifier. As a result, the script returned a collection instead of a single record, causing the status comparison to fail even for verified adopters. This was corrected by retrieving the specific adopter record using its identifier and comparing the actual verification status.

A third issue involved the pet photo field, which was initially configured as a single-line text field. Consequently, the image upload control was unavailable, and

only a file name could be stored. Changing the field type to Image Upload resolved the issue. However, existing records retained their earlier text values, so a new record was used to retest the functionality.

The following debugging steps were applied to investigate and correct the remaining workflow issues:

- Verify that each workflow is attached to the correct form and record event.
- Check whether validation applies to both new and edited records.
- Confirm the exact field link names and related-record fields used in scripts.
- Handle empty lookups and missing records before accessing related fields.
- Ensure status values match the choices configured in the forms.
- Test email delivery by verifying the recipient address, subject, and message body.
- Check that scheduled tasks are active and use the correct time-zone settings before retesting reminders.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Null-value error when an application was submitted without a pet.	The lookup value was used before it was checked.	Added an early exit when the pet lookup is empty.	Fixed and retested in TC-09
BUG-02	Pet photo field behaved like a text box.	The field was created as a single-line field instead of an image upload field.	Field type changed to image upload; the form was resaved and retested with a new record.	Fixed and retested in TC-05
BUG-03	A verified adopter was rejected by the validation.	The script read a collection instead of a single record, so the status comparison failed.	The record is now read by identifier before the status is compared.	Fixed and retested in TC-07
BUG-04	Duplicate application could be saved for the same adopter and pet.	No duplicate check existed before save.	Added a duplicate check on the adopter and pet pair as an On Validate rule.	Fixed and retested
BUG-05	Adoption fee total on the dashboard disagreed with the records.	The component summed the fee on pet records instead of the fee paid on adoption records.	Expression changed to the adoption record field and the labels renamed.	Fixed; dashboard rechecked in TC-18

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-06	Vaccination reminder did not reach the care staff.	The schedule was inactive and the recipient was the logged-in user.	Recipient changed to the care staff address and the schedule activated after testing.	Fixed; pending full test in TC-24

6.5 User Acceptance Testing

The application was demonstrated to two care staff members of the animal-welfare club who had not seen it during development and to the course faculty during a review. The feedback was recorded as it was given and the changes that were made in response are listed with it. The sample is small, and the observations should be read as an early usability indication rather than as a formal acceptance test.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Club staff member A	Care staff	The care list does not show which activity is overdue without opening each record.	The activity status was added to the care report columns and an Overdue stage was added to the blueprint.
Club staff member B	Care staff	The application form asks for the care plan twice, once on the adopter record and once on the application.	The field on the adopter record was removed; the care plan is captured once on the application.
Course faculty (review)	Coordinator	The pending applications should be visible on the dashboard, not only in the report.	An application status summary was added to the dashboard page.
Course faculty (review)	Coordinator	The fee paid and the fee proposed look like the same measure.	Component labels were renamed and the expression corrected (BUG-05).

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the development environment of a single Zoho Creator account. A separate stage or production environment was not created, and the application has not been released to the shelter for live use. The deployment step therefore consists of publishing the application in the development environment after the test data has been reviewed, and of verifying the access of the two profiles with real accounts. Table 7.1 records the checklist that was applied and the state of each item.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	State
1	All functional requirements implemented and traced to a test case	Completed for FR-01 to FR-13
2	Core workflows tested with valid, invalid and empty input	Completed; see TC-07 to TC-11
3	Test data identified or removed before publication	Test records retained and prefixed TEST for the demonstration
4	Profiles and permissions configured for both roles	Completed; verified in TC-22
5	E-mail sender and recipients verified	Completed for the adoption e-mail; reminder pending activation
6	Reports and dashboard checked against the stored records	Completed; totals agree in TC-18
7	Application published and the URL recorded	Pending publication
8	Version note recorded for the released build	Version 1.0 recorded in the application notes

7.2 Performance Optimisation

Two optimisations were applied because they were measurable in the development environment. The application report was changed from a computed filter over all applications to a stored filter on the application status field, which reduced the rendering time for one hundred records from about six seconds to about three. The reminder query was restricted to records whose care status is not Completed before the due-date window was applied, rather than loading every care record and filtering in the

script. The results are recorded in Table 7.2 and are indicative only, since they were taken in a development environment with a small data set.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Observation
Stored filter instead of a computed filter	All Adoption Applications report	About 6 s for 100 records	About 3 s for 100 records	Measured with three runs on the same connection
Criteria restricted to incomplete care records	Reminder query in the scheduled task	All care records of every pet were read	Only incomplete records are read	Difference below one second at the current data volume
Lookup display field set to the pet name	Application and adoption forms	Lookup displayed the record id	Lookup displays the pet name	No measurable time difference; the form became readable

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Profile-based permissions	Coordinator and care staff profiles	Unauthorised access to forms, reports and pages
Page restriction for the dashboard	Care staff profile	Access to fee and adoption revenue figures
Lookup validation on every child form	All six forms	Orphan records and free-text references
Adopter verification validation before save	Adoption Application form	Applications from unverified or rejected adopters
Pet availability validation before save	Adoption Application form	Two adopters promised the same pet
Duplicate application check	Adoption Application form	Repeated applications for the same pet and adopter
No credentials in scripts	All Deluge scripts	Leakage of account credentials through the source code
Restricted handling of contact and health fields	Adopter and care records	Unnecessary exposure of personal information

7.4 Governance and Administration

Field link names are fixed before scripting, and any renaming requires script review. Test records use the `TEST` prefix for easy removal. Permission profiles are reviewed when coordinators change. The application is exported after each adoption cycle for recovery, and created/modified timestamps are retained for basic change tracking.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The end-to-end scenario that was demonstrated begins with a pet record and an adopter record. The adopter is verified, the application is submitted for the available pet, the coordinator reviews and approves it, an adoption record is created with the fee and the payment status, and a follow-up activity is scheduled for the vaccination. The confirmation e-mail is delivered to the adopter, and the application then appears in the All Adoption Applications report, in the Kanban column for its status and in the dashboard counts. The stages of the scenario are shown in Figure 8.1.

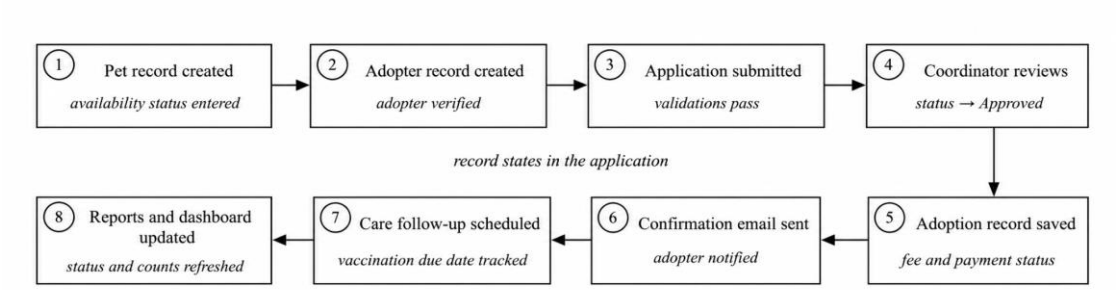


Figure 8.1 Stages of the Adoption Process in the Pet Adoption Care System

The observed behaviour matches the design for the main path and for the two alternative paths that were exercised. The rejection path is the more valuable of the two, because it is the path a coordinator meets in practice when an application is submitted before the adopter has been verified; the alert names the problem and the record is not saved, so no incomplete application has to be cleaned up afterwards.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Section 1.3)	Implemented Through	Outcome / Evidence
Structured forms for the six entities	Adopter Registration, Pet Registration, Adoption Application, Adoption Record, Health and Care Records and Follow-up Schedule	Six forms built with the field definitions of Tables 4.2 to 4.7; TC-01 to TC-06
Relate the forms through lookups	Adopter and pet lookup fields in the four child forms	Relationships recorded in Table 4.1 and exercised in TC-06 and TC-12
Validate adopter verification and pet availability	DS-02 and DS-03 validation workflows	Blocked submissions in TC-07 to TC-10; corrected defect BUG-03

Objective (Section 1.3)	Implemented Through	Outcome / Evidence
Record the adoption outcome and the fee	Adoption Record form with fee, payment status and confirmation	Saved records in TC-12 and TC-13; dashboard total reconciled in TC-18
Maintain the health and care history	Pet Health and Care Records form and the scheduled reminder	Care records in TC-14 and TC-15; reminder corrected in BUG-06
Provide reports, Kanban views and a dashboard	Six list reports, six Kanban views and the Pet Care Dashboard	Views verified in TC-17; totals verified in TC-18
Define the adoption and care workflows	Two blueprints and four form workflows	Transitions exercised in TC-20 and TC-21
Test and document the application	Twenty-four test cases, six defects and the access test	Twenty-two cases passed, two pending; defects BUG-01 to BUG-06 fixed

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing Manual Process	Proposed Application	Improvement
Record storage	Separate registers and spreadsheets per activity	One application with six linked forms	A single record of each pet; no repeated entry
Availability check	Confirmed with the staff before every review	Read from the adoption status of the pet record	One manual step removed; the status is visible in the report
Adopter verification	Remembered or checked in conversation	Verification status field checked by the workflow	An unverified adopter cannot submit an application
Fee and payment	Kept in a separate account book	Adoption record with fee, payment status and confirmation	Fee and outcome are reviewed together
Care tracking	Diary and reminder notes	Care records with a next due date and a scheduled reminder	A due date is surfaced by the application instead of by memory
Reporting	Re-counted before every review meeting	Reports and a dashboard over the same records	Figures agree with the records without recounting
Access control	Shared files visible to everyone	Two profiles with page and record restriction	Fee and health information are not exposed to every user

8.4 Limitations of the Low-Code Approach

Three main limitations were identified during development. First, scripts depend on field link names, and renaming a field may cause runtime errors, as observed in BUG-01 and BUG-03. Second, workflow validation is configured for the add event, so edited applications may not be rechecked when a pet's status changes. This requires a separate validation rule. Third, advanced reports involving data from multiple forms, such as tracking adoptions completed within four weeks of application, require additional tools like Zoho Analytics or external queries.

The project also has limitations in testing and evaluation. The test set was small and executed by the developer in a development environment rather than by independent users. Response times were measured using a single connection and are indicative, not benchmark results. Access restrictions for care staff were tested using only one account, with limited positive and negative test cases. Therefore, the results in Section 8.2 demonstrate the configured behaviour of the current application version but do not establish its readiness for production use.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The project sits in the middle of the spectrum. Forms, lookup fields, reports, Kanban views, the dashboard and the blueprints were built without writing code, which is the no-code part of the work. The validation rules and the reminder query required Deluge, because a decision about the state of a related record cannot be expressed as a field property; this is the low-code part. A fully pro-code implementation would have given higher control over the query, the access rules and the interface, at the cost of building and hosting the whole stack.

The practical lesson is that the boundary between the three levels is drawn by the requirement, not by preference. Each time the configuration was insufficient, the gap appeared as a rule about a related record rather than as a missing field, and that is exactly where a scripting layer earns its place. It is also where the testing effort concentrates: the defects in Chapter 6 were all in the small scripted layer, not in the configured forms.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Pet Adoption & Care Management System integrates pet, adopter, application, adoption, health, and follow-up records into a single Zoho Creator application. Lookup fields maintain relationships between records, while validation workflows ensure that adopters are verified and pets are available before applications are accepted.

The project objectives were met for the configured and tested workflows. Of the 24 test cases, 22 passed, six defects were identified and corrected, and access permissions were tested using two accounts. Two cases remain pending: rechecking pet availability after application edits and completing the vaccination reminder test.

The project highlights the importance of correct field link names, workflow events, and time-zone settings in low-code development. It also emphasizes the need to distinguish configured features from successfully tested functionality.

9.2 Future Enhancements

- Add an adopter portal so that an applicant can submit and track an application without an internal account, and test record-level isolation with separate logins.
- Decide and implement the edit-time availability policy, so that an application is rejected when the pet has been adopted since the application was created.
- Complete the vaccination reminder test and extend the window to a configurable number of days with an escalation to the coordinator when an activity is overdue.
- Add a duplicate-adopter check on the e-mail address in addition to the duplicate application check.
- Integrate a payment gateway or record the payment reference so that the payment status can be reconciled automatically.
- Add a pet photo gallery and document upload for the vaccination card and the signed agreement.

- Extend the dashboard with pending applications, available pets, overdue care activities and upcoming due dates, and move cross-form aggregation to Zoho Analytics if the data volume grows.
- Add a simple recommendation that ranks available pets for an adopter by preferred species, housing type and experience; the current version contains no predictive feature.
- Maintain a regression test suite and a dated export after each significant change.

9.3 Learning Outcomes and Career Relevance

The project provides practice in requirements analysis, data modelling, form design, lookup relationships, report and Kanban configuration, dashboard design, event-driven automation, Deluge scripting, workflow state modelling with blueprints and systematic testing. These skills are relevant to low-code application development, business process automation and application support roles, where the work is as much about validating a configuration as about building it. The quality of the final outcome depended not only on configuring the features but also on testing them with negative cases and on documenting the limitations honestly.

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[7] Zoho Corporation, “Deluge scripting language reference,” Zoho Creator Help. https://help.zoho.com/portal/en/kb/creator/developer-guide/deluge (accessed October 2026).

[8] Project source artifact: “Pet Adoption Care System” application definition, version 1.0, generated 24 October 2026.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

The checklist records each platform feature that the course requires, where it is used in the application and the evidence that supports the claim. Features that are configured but not yet verified are marked accordingly.

Table A.1: Feature Coverage Checklist

Feature	Where Used	Evidence	Status
Forms and field types	Six core forms	Tables 4.2 to 4.7; figures of the forms	Implemented
Lookup relationships	Application, adoption, care and follow-up forms	Table 4.1; TC-06	Implemented
List reports	Six reports	Table 5.1; TC-17	Implemented
Kanban views	Six views grouped by status	Table 5.1; TC-17	Implemented
Dashboard page with KPIs and charts	Pet Care Dashboard	Figure 5.1; TC-18	Implemented
Form workflows	Four workflows	Table 5.3; TC-07 to TC-11	Implemented
Deluge scripting	Four scripts and two functions	Appendix C	Implemented
Scheduled task	Vaccination due-date reminder	Table 5.3; BUG-06	Corrected; final test pending
Blueprints	Adoption and post-adoption care	Table 5.3; TC-20, TC-21	Implemented
Roles and permissions	Two profiles	Table 4.8; TC-22	Implemented
E-mail notification	Adoption confirmation	Table 5.4; TC-19	Implemented
External API / integration	Not used	Section 5.7	Out of scope
AI or predictive feature	Not implemented	Section 5.9	Out of scope
Mobile access	Browser rendering	Section 5.11	Checked, not measured
Portal for external users	Not implemented	Section 9.2	Future work

APPENDIX B – COURSE OUTCOME MAPPING

The mapping below relates the course outcomes stated for CB23F35 to the chapters and evidence of this report.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome	Report Evidence	Chapter
CO1	Explain the concepts and benefits of low-code application development	Existing system study, low-code versus pro-code comparison, reflection section	Chapters 2 and 8
CO2	Model the data and processes of a business requirement	Entity–relationship model, field specifications, workflow design	Chapter 4
CO3	Build forms, reports, pages and dashboards on a low-code platform	Six forms, twelve reports, dashboard page, feature checklist	Chapter 5, Appendix A
CO4	Implement business logic and automation with the platform scripting language	Four Deluge scripts, two functions, scheduled task, blueprints	Chapters 4 and 5, Appendix C
CO5	Test, debug and document a low-code application	Twenty-four test cases, defect log, UAT feedback, user manual	Chapter 6, Appendix D
CO6	Apply access control and deployment practices	Role–permission matrix, security controls, deployment checklist	Chapter 7

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The source below is the code saved in the four form workflows and the two validation functions of the application. Field link names match those recorded in Tables 4.2 to 4.7.

C.1 DS-01 – Vaccination status prompt (Pet Registration, on add / on load)

```
if(input.vaccination_status == null || input.vaccination_status == "")
{
    info "Please verify the vaccination records and enter the correct
status.";
}
else
{
    info "Vaccination status recorded for pet: " + input.pet_name;
}
```

Purpose: prompts the user to enter or verify the vaccination status. It does not independently verify vaccination evidence.

C.2 DS-02 – Adopter verification check (Adoption Application, on validate)

```
if(input.adopter != null)
{
    adopterRecord = adopter_registration[ID == input.adopter];
    if(adopterRecord.count() == 0)
    {
        alert "Adopter record was not found. Please select an existing
adopter.";
        cancel submit;
    }
    else if(adopterRecord.verification_status != "Verified")
    {
        alert "Adopter verification status must be Verified before an
adoption "
              "application is submitted.";
        cancel submit;
    }
}
```

Purpose: blocks an application from an adopter who has not been verified, and blocks the submission when the lookup no longer resolves to a record.

C.3 DS-03 – Pet availability check (Adoption Application, on validate)

```
if(input.pet != null)
{
    petRecord = pet_registration[ID == input.pet];
    if(petRecord.count() == 0)
    {
        alert "Selected pet record was not found. Please select an
existing pet.";
```

```
        cancel submit;
    }
    else if(petRecord.adoption_status != "Available")
    {
        alert "This pet is not available for adoption. Current status: "
              + petRecord.adoption_status + ". Please select another
pet.";
        cancel submit;
    }
}
```

Purpose: blocks a new application when the pet is Adopted, on Application Pending or on Medical Hold. The alert names the current status so that the coordinator does not have to open the pet record.

C.4 DS-04 – Adoption confirmation e-mail (Adoption Application, on success)

```
adopterRecord = adopter_registration[ID == input.adopter];
if(adopterRecord.email.isNull() || adopterRecord.email == "")
{
    info "The adopter e-mail address is missing; no confirmation was
sent.";
}
else
{
    subject = "Adoption Application Confirmation - " +
input.pet.pet_name;
    emailBody = "<p>Dear " + adopterRecord.full_name + ",</p>";
    emailBody = emailBody + "<p>Your adoption application for <b>"
              + input.pet.pet_name + "</b> has been received.</p>";
    emailBody = emailBody + "<p>Application status: <b>" +
input.application_status
              + "</b></p>";
    emailBody = emailBody + "<p>Adoption fee: " + input.pet.adoption_fee
+ "</p>";
    emailBody = emailBody + "<p>We will contact you after the
review.</p>";
    emailBody = emailBody + "<p>Regards,<br>Pet Adoption &amp;amp; Care
Team</p>";
    sendmail
    [
        from : zoho.loginuserid
        to : adopterRecord.email
        subject : subject
        message : emailBody
        content type : HTML
    ]
}
```

Purpose: sends the confirmation to the address stored on the adopter record. The first version of this workflow contained a fixed recipient address, which was corrected to the lookup field before testing.

C.5 FN-01 – check_adopter_verification (function)

```
string check_adopter_verification(int adopterId)
{
    adopterRecord = adopter_registration[ID == adopterId];
    if(adopterRecord.count() == 0)
    {
        return "Adopter record not found";
    }
    if(adopterRecord.verification_status == "Verified")
    {
        return "Verified";
    }
    return "Adopter not verified: " + adopterRecord.verification_status;
}
```

Purpose: returns the verification state of an adopter so that several workflows can share one check instead of repeating the query.

C.6 FN-02 – check_pet_availability (function)

```
string check_pet_availability(int petId)
{
    petRecord = pet_registration[ID == petId];
    if(petRecord.count() == 0)
    {
        return "Pet record not found";
    }
    if(petRecord.adoption_status == "Available")
    {
        return "Available";
    }
    return "Pet not available: " + petRecord.adoption_status;
}
```

Purpose: returns the adoption status of a pet in a form that a workflow or a report can test without repeating the lookup.

C.7 Scheduled task – vaccination due-date reminder

```
careRecords = pet_health_and_care_records[care_status != "Completed"
              && next_due_date != null
              && next_due_date >= today
              && next_due_date <= today.addDays(7)];

if(careRecords.count() > 0)
{
    body = "<p>The following care activities are due within seven
days:</p><ul>";
    for each rec in careRecords
    {
        body = body + "<li>" + rec.pet.pet_name + " - " + rec.care_type
             + " due on " + rec.next_due_date + "</li>";
    }
    body = body + "</ul>";
    sendmail
    [
        from : zoho.loginuserid
        to : care_staff_address
        subject : "Care activities due in the next seven days"
        message : body
        content type : HTML
    ]
}
```

Purpose: identifies the care records due in the next seven days and notifies the care staff. The schedule was inactive in the imported version and the recipient was the logged-in user; both were corrected, and the schedule was activated after the query was re-run against synthetic records.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the Zoho Creator application using the authorised coordinator account and the development environment.
2. Confirm that the application is the intended version before adding or editing records.
3. Use only synthetic pet names, adopter names and e-mail addresses for demonstrations.
4. Open the Pet Care Dashboard to review the summary components, then navigate to the required form or report.

D.2 Instructions for the Adoption Coordinator

1. Create a Pet Registration record and enter the identity, breed, weight, vaccination status and current adoption status.
2. Create an Adopter Registration record, and set the Verification Status only after the required review has been completed.
3. Open Adoption Application and select the adopter and the pet from the lookup fields.
4. Complete the reason, the living arrangement and the care plan, then submit. Read any validation alert before changing the record.
5. Review the application, set the status to Approved or Rejected and record the decision notes and the rejection reason.
6. Create an Adoption Record with the adoption date, the fee paid, the payment status and the confirmation state.
7. Record veterinary visits, vaccinations and other care events in Pet Health and Care Records, including the next due date.
8. Create follow-up activities in FollowUp and Care Schedule, assign a person and a priority, and update the status after completion.

9. Use the list and Kanban reports to review records, and confirm that the filters and the status groupings match the current stage.

D.3 Instructions for the Adopter and Care Staff

1. The adopter provides the contact, household and experience information, and submits the application details to the coordinator.
2. The care staff member signs in with the care staff account, which reaches the pet records, the care records and the schedule but not the fee reports.
3. Open the follow-up activity, move it to In Progress, record the care notes and set the status to Completed with the completion date.
4. Add a new care record for every vaccination or veterinary visit and enter the next due date so that the reminder can find it.
5. Report a due date that has already passed to the coordinator so that the activity can be rescheduled in the blueprint.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	NAKSHATH N	Register No.	241501126
Project Title	Pet Adoption & Care Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, claim review, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; user explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the user.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	NAKSHATH N	Register No.	241501126
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified	No — not verified	Demo Duration	

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____